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Using Lagrange multiplier type tests to detect structural intra-person heterogeneity in composite marginal likelihood estimation in panel data sets☆

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ABSTRACT

Gradient-based Lagrange multiplier-type tests represent a valuable tool for discriminating between nested models, obviating the necessity to estimate the unrestricted model. This is particularly advantageous when testing for pooling in panel data sets, as it permits the testing of multiple groupings without the necessity of re-estimating the model for each grouping. This makes the process considerably faster and more flexible in comparison to Wald or likelihood ratio type tests.

In this paper, we demonstrate that the use of pairwise composite marginal likelihood (CML) estimation enables the comparison of gradients between different CML contributions of pairs of observations for individuals. This allows for the testing of pooling over time, as well as the identification of neglected temporal correlation. The CML approach thus offers a degree of flexibility that is not present in the classical likelihood setting.

Theoretical derivations of the asymptotic distribution of the test statistics under the null hypothesis are provided for the special case of multinomial probit models, thereby forming the basis for the statistical interpretation of the test statistic.

Moreover, a comprehensive simulation study was conducted to assess the finite-sample performance of the test statistics. In particular, the distribution of the test statistic under the null hypothesis and the rejection rates of the tests under various types and degrees of violations of the null hypothesis were evaluated using synthetic panel data sets of varying sizes. This empirical evaluation provides insights into the effectiveness and reliability of the proposed tests in detecting intra-personal heterogeneity and into causes of misspecifications in the deterministic utility structure.

1. Introduction

Multinomial probit (MNP) models are used for modelling discrete choices made by decision-makers, with the understanding that these choices depend on a set of covariates which describe the characteristics of the decision-maker in question, as well as the specific choice alternatives. In accordance with the random utility interpretation of these models, it is assumed that decision-maker n forms an utility $U_{n,t,j}$ for each alternative $j \in \{1, \dots, J\}$ in their tth choice situation, where $J \in \mathbb{N}$ denotes the number of possible alternatives. This utility is composed of a systematic part $V_{n,t,j}$ and a random (unobservable) part $\varepsilon_{n,t,j}$. The decision-maker then

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selects the alternative that provides the highest utility. MNP models assume additive separability and normal distribution for the error term, such that the utility can be expressed as

$$U_{n,t,j} = V_{n,t,j} + \varepsilon_{n,t,j}, \quad \varepsilon_{n,t,.} = (\varepsilon_{n,t,j})_{j=1,\dots,J} \in \mathbb{R}^J, \quad \varepsilon_{n,t,.} \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $\mathcal{N}(0, \Sigma)$ denotes the multivariate normal distribution with expectation zero and variance $\Sigma \in \mathbb{R}^{J\times J}$. It is then assumed that the observed choice $y_{n,t}$ represents the alternative with the highest random utility, therefore $y_{n,t} = \arg\max_j U_{n,t,j}$. Such models have been used extensively in a range of disciplines, including transportation (Bhat and Koppelman, 2003; Büscher et al., 2019), marketing (see, for example, Albert and Chib, 1993; Chintagunta, 1992), and psychology (Johnson and Bruce, 1997; Berkowitsch et al., 2014), to mention just a few sources.

In many cases it may be questioned whether preferences (as encoded in the systematic part of the utility) are homogeneous across decision-makers. In addition, the assumption that preferences remain stable over time may warrant further investigation. The random term of the utility, nevertheless, is typically assumed to be independent, identically distributed (iid) over subsequent decisions, ignoring behaviour such as variety seeking (leading to negative autocorrelation) or habit formation (related to positive autocorrelation).

In certain instances, the heterogeneity of decision-makers can be directly modelled by incorporating observed characteristics of the decision-makers into the model. One illustrative example in this regard is the incorporation of income as a factor influencing purchase decisions. Other forms of preference heterogeneity have been addressed through the utilisation of fixed effects in panel data scenarios with a large number of repeated choices for relatively few decision-makers (small N, large T panels). More often, random effects are used in large N small T panels, assuming that the preference heterogeneity is independent of the other regressors.

It is evident that, when starting with an initial homogeneous model, there are numerous potential deviations that may need to be taken into account:

(I) unobserved heterogeneity of decision-makers,
(II) changes in preferences over time,
(III) autocorrelation in the random term.

In order to detect these deviations, it is necessary to employ diagnostic instruments. In many instances, the diagnostic assessment entails selecting between a more parsimonious model and an extension within a nested framework, such that the simpler model represents a restriction of the extended model. To discriminate between an unrestricted and a linearly restricted model, in the context of maximum likelihood estimation, one can choose from the classical triad of tests (Silvey, 1959):

- A Wald-Test (Wald, 1943), where the standardised distance from the unrestricted parameter estimate $\hat{\theta}_{\text{ur}} \in \mathbb{R}^P$ to the restriction $R\theta = r \in \mathbb{R}^k$, with $R \in \mathbb{R}^{k\times P}$ is assessed using the test statistic $W = (R\hat{\theta}_{\text{ur}} - \text{r})'(R\hat{V}_{\hat{\theta}_{\text{ur}}}R')^{-1}(R\hat{\theta}_{\text{ur}} - \text{r}) \overset{a}{\sim} \chi^2_k$ under the null hypothesis $H_0 : R\theta = r$, where k denotes the number of restrictions and $V_{\hat{\theta}_{\text{ur}}} \in \mathbb{R}^{P\times P}$ the estimated covariance matrix of the maximum likelihood estimator of the unrestricted model.
- The Likelihood-Ratio (LR) test (Wilks, 1938), is used to compare the ratio between the maximum likelihood values of the restricted and the unrestricted model. More specifically, the test statistic consists of the natural logarithm of the ratio with $LR = -2\log(L(\hat{\theta}_{\text{r}})/L(\hat{\theta}_{\text{ur}})) \overset{a}{\sim} \chi^2_k$ under the null hypothesis, where $L(\hat{\theta})$ denotes the likelihood function evaluated at $\hat{\theta}$ and $\hat{\theta}_{\text{r}} \in \mathbb{R}^P$ denotes the estimator in the restricted model.
- The Lagrange multiplier (LM) test (Breusch and Pagan, 1980), also referred to as score test, assesses the gradient of the unrestricted model at the maximum likelihood parameter estimate of the restricted model

$$LM = \left(\partial_{\theta_{\text{ur}}} \log L(\hat{\theta}_{\text{r}})\right)' I(\hat{\theta}_{\text{r}})^{-1} \left(\partial_{\theta_{\text{ur}}} \log L(\hat{\theta}_{\text{r}})\right) \overset{a}{\sim} \chi^2_k, \tag{2}$$

  with $I(\hat{\theta}_{\text{r}})$ denoting an estimator of the Fisher information matrix $-\mathbb{E}\left(\partial_{\theta_{\text{ur}}}\partial_{\theta'_{\text{ur}}} \log L(\hat{\theta}_{\text{r}})\right)$.

Among these three tests, which are asymptotically equivalent, the LM test has the advantage that it requires only the estimation of the restricted model, as the gradient and Hessian with respect to the unrestricted parameter vector θ_{ur} are evaluated at the restricted maximum likelihood estimate $\hat{\theta}_{\text{r}}$. This is of particular pertinence for models for which estimation times are not negligible, as is frequently the case for discrete choice models.

The aforementioned deviations (I)–(III) from the restricted model correspond to changes in the parameter vector, either over decision-makers (in case of unobserved heterogeneity), over decisions taken (for example, for structural breaks in preferences over time), or to nonzero correlations of the error terms.

In order to illustrate the main idea of this paper, consider a standard linear panel data model with standard normally distributed error terms:

$$y_{n,t} = X'_{n,t}\beta + u_{n,t}, \quad n = 1, \dots, N,\ t = 1, \dots, T,\ u_{n,t} \sim \mathcal{N}(0,1). \tag{3}$$

The logarithm of the Gaussian likelihood and the score function for this model can be written as (using shorthand notation for the data set)

$$ll(\beta; y, X) = c - 1/2 \sum_{n=1}^{N} \sum_{t=1}^{T} (y_{n,t} - X'_{n,t}\beta)^2, \tag{4}$$

$$\partial_\beta ll(\beta; y, X) = \sum_{n=1}^{N} \sum_{t=1}^{T} X_{n,t}(y_{n,t} - X'_{n,t}\beta), \tag{5}$$

from which it follows that at the maximum likelihood estimate $\hat{\beta}$

$$0 = \partial_\beta ll(\hat{\beta}; y, X) = \sum_{n=1}^{N} \sum_{t=1}^{T} X_{n,t}(\underbrace{y_{n,t} - X'_{n,t}\hat{\beta}}_{\hat{u}_{n,t}}), \tag{6}$$

and thus $\hat{\beta} = (\sum_{n=1}^{N} \sum_{t=1}^{T} X_{n,t} X'_{n,t})^{-1} \sum_{n=1}^{N} \sum_{t=1}^{T} X_{n,t} y_{n,t}$.

Let us now assume that there is reason to believe that the set of individuals can be partitioned into two groups, designated G_1 and G_2, where the parameter vector β differs between the groups. In this case, the model would be extended to

$$y_{n,t} = X'_{n,t}\beta + X'_{n,t}\beta_2 z_n + u_{n,t}, \quad n = 1, \dots, N,\ t = 1, \dots, T,\ u_{n,t} \sim \mathcal{N}(0,1), \tag{7}$$

where z_n indicates whether the individual n is in group G_2. The restricted model corresponds to $\beta_2 = 0$, while $\beta_2 \neq 0$ implies heterogeneous parameters between the groups. The score of the extended unrestricted model is comprised of two elements: the zero derivative provided above, and the derivative with respect to β_2, which is equal to

$$\sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t}(y_{n,t} - X'_{n,t}\hat{\beta}) = \sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t} y_{n,t} - \left(\sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t} X'_{n,t} \right) \hat{\beta}, \tag{8}$$

where $\tilde{z}_n = z_n - 1/2$. Now assume that the population is matched, that is to say, for each individual n_1 in group $G_1 = \{1, 2, \dots, N/2\}$, there exists an individual $n_2 = N/2 + n_1$ in group $G_2 = \{N/2 + 1, \dots, N\}$, such that

$$\sum_{t=1}^{T} X_{n_1,t} X'_{n_1,t} = \sum_{t=1}^{T} X_{n_2,t} X'_{n_2,t}, \tag{9}$$

implying $\sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t} X'_{n,t} = 0$, which in turn means that the score with respect to β_2 equals

$$\begin{aligned} \sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t} y_{n,t} &= \sum_{n=1}^{N} \sum_{t=1}^{T} \tilde{z}_n X_{n,t} \hat{u}_{n,t} \\ &= 1/2 \sum_{n=1}^{N/2} \underbrace{\left(\sum_{t=1}^{T} X_{n+N/2,t} \hat{u}_{n+N/2,t} - \sum_{t=1}^{T} X_{n,t} \hat{u}_{n,t} \right)}_{\Delta \bar{g}_n}. \end{aligned} \tag{10}$$

Under the null hypothesis, it follows that $\mathbb{E}\Delta\bar{g}_n = 0$ and the score is asymptotically normally distributed. Its variance can be estimated from $\Delta\bar{g}_n$, given the assumption of iid sampling over individuals. This provides the foundation for the construction of an LM test statistic, which can be used to investigate the potential for pooling the data in this straightforward scenario.

In this case, it is evident that the roles of N and T can be switched to compile an LM test statistic for testing a structural break in time using the test statistic

$$LM = 1/2 \sum_{t=1}^{T/2} \underbrace{\left(\sum_{n=1}^{N} X_{n,t+T/2} \hat{u}_{n,t+T/2} - \sum_{n=1}^{N} X_{n,t} \hat{u}_{n,t} \right)}_{\Delta \bar{g}_t}. \tag{11}$$

Both statistics primarily utilise the gradient $X_{n,t}\hat{u}_{n,t}$ of the log-likelihood contribution of each individual at each time point, taking differences between different time points or different groups (depending on the effect under investigation) in conjunction with the assumption of iid sampling over these dimensions.

This concept is extended in this paper for MNP models in a panel structure with random effects, with the aim of detecting unobserved heterogeneity, structural breaks in time, as well as temporal autocorrelation in the random component. In order to generalise the argument, it should be noted that the systematic part of the utility is most often assumed to be linear and analogous to the linear model above, given as $V_{n,t,j} = \beta' X_{n,t,j}$. It is evident that the extension to the unrestricted model

$$V_{n,t,j} = \beta' X_{n,t,j} + z_n \beta'_2 X_{n,t,j} \tag{12}$$

is appropriate when considering two groups of decision-makers with different preferences. This is then typically estimated by maximising the log-likelihood function

$$\begin{aligned} ll_{\mathrm{ML}}(\theta; y, X) &= \sum_{n=1}^{N} \log \Pr(U_{n,1,y_{n,1}} \geq U_{n,1,i} \ \forall i, \ldots, U_{n,T,y_{n,T}} \geq U_{n,T,i} \ \forall i \mid y_n, X_n) \\ &= \sum_{n=1}^{N} \log L_n(\theta). \end{aligned} \tag{13}$$

Assuming conditional independence of the observations across time, conditional on observed covariates, this simplifies to

$$ll_{\mathrm{ML}}(\theta; y, X) = \sum_{n=1}^{N} \sum_{t=1}^{T} \log \Pr(U_{n,t,y_{n,t}} \geq U_{n,t,i} \ \forall i \mid y_{n,t}, X_{n,t}) = \sum_{n=1}^{N} \sum_{t=1}^{T} \log L_{n,t}(\theta). \tag{14}$$

However, mixing, for example due to random coefficients β_n in the systematic part $\beta_n' X_{n,t,j}$ of the utility (where β_n is drawn iid from some underlying distribution independent of all observed $X_{n,t,j}$), implies that the log-likelihood contribution for decision-maker n corresponding to the mixed MNP cannot be written as a sum of the terms for different choice occasions t, but only within the joint term $L_n(\theta)$. This limits the flexibility since a switching of the roles of T and N is no longer possible.

Using composite marginal likelihood (CML) instead of maximum likelihood estimation entails the replacement of the joint log-likelihood of all observations $\log L_n(\theta)$ of an individual n, with a weighted sum of the log-likelihoods of margins. In the following, as is typically the case, pairs of observations are considered for the margins, resulting in the (pairwise) CML function

$$\begin{aligned} ll_{\mathrm{CML}}(\theta; y, X) &= \sum_{n=1}^{N} \sum_{a=1}^{T_n-1} \sum_{b=a+1}^{T_n} w_{n,a,b} \log \Pr(U_{n,a,y_{n,a}} \geq U_{n,a,i}, \\ &\qquad U_{n,b,y_{n,b}} \geq U_{n,b,i} \ \forall i \mid y_{n,a}, y_{n,b}, X_{n,a}, X_{n,b}) \\ &= \sum_{n=1}^{N} \sum_{a=1}^{T_n-1} \sum_{b=a+1}^{T_n} w_{n,a,b} \log L_{n,a,b}(\theta), \end{aligned} \tag{15}$$

where T_n can vary over different individuals, allowing for unbalanced panel settings. The second equation defines the term $L_{n,a,b}(\theta)$. Analogously to the linear model case previously discussed, a number of different groupings can be used to provide different LM tests. The application of CML estimation permits the extension of LM pooling tests from the individual level to the level of margins. This flexibility allows for the investigation of unaccounted temporal effects and intra-individual heterogeneity without the necessity of incorporating these effects into the model. However, for valid inference, it is essential to recognise that pairs of observations from a single individual are not independent, as the same choice occasion may be included in multiple pairs of observations.

Consequently, the correlation between gradient contributions within decision-makers can differ from the correlation across different decision-makers. The proposed LM test thus first defines groups of pairs of observations, subsequently calculates the mean across the intra-decision-maker contributions for each group, and finally estimates the variance between the contributions from different individuals. This process serves to relax the assumptions for the test.

It is also important to note that, in the context of CML, the analogue to likelihood ratio tests given by log-CML ratios does not lead to nuisance parameter free tests, contrary to the likelihood ratios. Rather, under the null hypothesis, the log-CML ratio tends asymptotically to a weighted χ^2 distribution, where the weights depend on the product of the inverse of the Hessian matrix with the variance of the score (compare Varin et al., 2011). The same issue is also prevalent in information criteria such as the AIC, which must be adapted to the CML situation and demonstrate worse performance in simulations in comparison to t-tests for model selection (see, for example, Batram, 2023). The calculation of Wald-type tests based on CML estimates requires the estimation of both terms, the Hessian and the variance of the score, as well as the estimation of the unrestricted model. This adds a considerable amount of calculations (see below), making this test rather unattractive.

In this paper, we present a new Lagrange multiplier-type test, capable of testing for structural intra-person heterogeneity in CML estimation. In Section 2, we present a gradient-based test, similar to the Lagrange multiplier test, that accounts for the fact that pairs of observations from one individual are not independent. Furthermore, we derive the asymptotic distribution of the test statistic. In Section 3, we employ the discrete choice modelling setting, specifically MNP models, to generate synthetic data to demonstrate the finite sample properties of the proposed test procedure. MNP models are particularly well-suited to this purpose, given that they are computationally demanding such that CML estimation and its advancements have become popular methods for estimating these models (see, for instance, Bhat, 2011; Varin, 2008; Varin et al., 2011). In Section 3.1, we first confirm the theoretical distribution of the test statistic under the null hypothesis for finite samples. In Sections 3.2 and 3.3, we then proceed to implement violations of the null hypothesis of different types and degrees of severity in the data generating process (DGP) and evaluate the power properties of the test in these cases. The results of these simulations are presented in Section 4, and conclusions regarding the applicability of the presented testing method are presented in Section 5.

The R and C++ code used for the estimation processes in this paper is bundled into an R-package named *Rprobit*, which is available on https://github.com/dbauer72/Rprobit.

2. CML pair gradient contributions

The core of this paper lies in the understanding of the score of the CML function and how it is composed of the contributions of the individual CML margins of all decision-makers. This section is therefore dedicated to understanding and dissecting the score function of the CML function, analysing the distribution of its components and then combining the components to form LM type tests.

The score corresponding to the CML function (15) can be expressed as

$$\partial_\theta ll_{CML}(\theta; y, X) = \sum_{n=1}^{N} \sum_{a=1}^{T_n-1} \sum_{b=a+1}^{T_n} w_{n,a,b} \partial_\theta \log L_{n,a,b}(\theta). \tag{16}$$

When calculating the CML estimator, $ll_{CML}(\theta; y, X)$ is maximised w.r.t. θ, without restrictions, and thus the score (16) at the optimum $\hat{\theta}$ is equal to zero. The aforementioned score is composed of the sum of the gradient contributions of the individual CML pairs of observations

$$g_{n,a,b}(\theta) = \partial_\theta \log L_{n,a,b}(\theta). \tag{17}$$

In the context of an exchangeable MNP (under the null hypothesis H_0), whereby all pairs of observations are assumed to follow the same distribution, it can be shown that $\mathbb{E}(g_{n,a,b}(\theta_\circ)) = 0$, where $\theta_\circ$ denotes the true parameter vector of the DGP. However, unobserved heterogeneities, structural breaks in time, as well as temporal autocorrelation in the random terms (deviations (I)–(III)) all lead to systematic differences in the joint choice probabilities for pairs of choices that can be detected from the log-CML gradient contributions $g_{n,a,b}(\hat{\theta})$.

This paper's primary result discusses the properties of $g_{n,a,b}(\hat{\theta})$ under a set of assumptions regarding the DGP and the CML function.

Assumption 1 (*Data Generating Process*)**.** The data set $(y_n, X_n), n = 1, \dots, N$ is generated by the following mechanism:

(I) A number T_n of choice occasions are drawn from a discrete random distribution, supported in the set $\{2, 3, \dots, \bar{T}\}$, and are independent of covariate values. In this context, $\bar{T} \in \mathbb{N}_{>1}$ denotes the maximum number of observed choices for one individual.

(II) For each decision-maker facing T_n choice occasions, a matrix $X_n = [X_{n,1,.}, \dots, X_{n,T_n,.}] \in \mathbb{R}^{PJ \times T_n}$ of regressors is chosen iid across decision-makers, such that for each pair of choice occasions (a, b), the matrix $[X_{n,a,.}, X_{n,b,.}]$ is distributed identically over pairs $(a, b), 1 \le a < b \le T_n$. Furthermore, $\|X_{n,a,.}\|_\infty \le M, 1 \le a \le T_n$ for some scalar $M < \infty$ (uniform norm bound).

(III) For given T_n and X_n, the vector of choices $y_n = [y_{n,1}, \dots, y_{n,T_n}]' \in \mathcal{J}^{T_n}, \mathcal{J} = \{1, \dots, J\}$ is chosen according to the mixed MNP model with normally distributed random effects corresponding to parameter vector $\theta_0 \in \mathbb{R}^P$ for appropriate integer P.

(IV) The regressors X are chosen from a distribution in a manner that ensures that the model parameters are locally identifiable (i.e. no multicollinearity between regressors and the model is identified w.r.t. the scale and level of the utilities).

Assumption 2 (*CML Weights*)**.** The CML weights $w_{n,a,b}$ are chosen according to one of the following five schemes:

(I) $w_{n,a,b} = f(a, b)$ for some bounded positive function $f : \mathbb{N}^2 \to [\underline{w}, \overline{w}]$, $0 \le \underline{w} \le \overline{w} < \infty$.

(II) $w_{n,a,b} = \mathring{w}_{T_n} \in [\underline{w}, \overline{w}]$, $0 < \underline{w} \le \overline{w} < \infty$ (*groupwise* CML weights).

(III) $w_{n,a,b} \in \{0, 1\}$ chosen randomly independent of all other variables, iid over decision-makers, such that

$$\sum_{a=1}^{T_n-1} \sum_{b=a+1}^{T_n} w_{n,a,b} = w_{T_n} \tag{18}$$

(selecting w_{T_n} random pairs from within $T_n(T_n - 1)/2$ possible pairs).

(IV) Stratification weights $w_{n,a,b} = w_n$ drawn iid over decision-makers from some underlying distribution supported on $[\underline{w}, \overline{w}]$, $0 < \underline{w} \le \overline{w} < \infty$.

(V) A combination of (I)–(IV).

Under these assumptions, the following theorem can be obtained (for the proof of the theorem, see Appendix A):

Theorem 1. *Let the data be generated from a mixed MNP model with utility* $U_{n,t,j} = X'_{n,t,j}\beta + X'_{n,t,j}\gamma_n + \varepsilon_{n,t,j}$ *for* $n = 1, \dots, N$, $t = 1, \dots, T_n$, *where* $\gamma_n \sim \mathcal{N}(0, \Omega)$ *independent of* $X_{n,t,j}$, *and* $\varepsilon_{n,t,j}$ *iid over decision-makers, such that Assumption 1 is satisfied.*

Let the model be estimated using CML with weights $w_{n,a,b}$ *in accordance with Assumption 2, providing the estimator* $\hat{\theta}$ *maximising the CML such that the score is zero and the estimator is asymptotically normal with* $\sqrt{N}(\hat{\theta} - \theta_\circ) \xrightarrow{d} Z \sim \mathcal{N}(0, V_\theta)$.

Then

$$g_{n,a,b}(\hat{\theta}) = \partial_\theta \log L_{n,a,b}(\hat{\theta}) = g_{n,a,b}(\theta_\circ) + \partial_\theta g_{n,a,b}(\bar{\theta}_n)(\hat{\theta} - \theta_\circ), \tag{19}$$

where $\mathbb{E}(g_{n,a,b}(\theta_\circ)) = 0$. *Furthermore,* $\mathbb{E}\partial_\theta g_{n,a,b}(\theta_\circ) = H_g > 0$ *and* $\bar{\theta}_n$ *is an intermediate value such that* $\bar{\theta}_n \xrightarrow{p} \theta_\circ$ *as* $N \to \infty$.

Finally,

$$N^{-1}\sum_{n=1}^{N}\partial_\theta g_{n,a,b}(\bar{\theta}_n)(\hat{\theta}-\theta_\circ) = H_g(\hat{\theta}-\theta_\circ) + o_P(N^{-1/2}). \tag{20}$$

The theorem demonstrates that the score contribution of each pair is essentially equal to the score contribution evaluated at the true parameter vector, plus an additional term that depends on the estimation error multiplied by the Hessian, which does not depend on the individual.

The aforementioned assumptions imply that pairs can be used interchangeably, as they are based on regressors drawn from the same underlying stochastic mechanism as are the random parts of the utility. This is, of course, a strong assumption, which is mostly adequate in panel studies with repeated choices of the same decision problem.

It should be noted, however, that the MNP model is employed here merely for illustrative purposes. The theorem does not utilise this structure and is also applicable for multinomial logit models or any other model class, provided that pairs of observations employ the same underlying conditional (on the regressors) and unconditional likelihoods.

The theorem provides us with the means to use for diagnostic checking: In the context of the restricted model, all pairs of choice occasions are identically distributed. In particular, the corresponding contribution to the score, apart from a common term accounting for the estimation error, consists of the score contribution evaluated at the true parameter vector, which has mean zero. If the population, however, can be partitioned into two groups corresponding to different parameter vectors generating the data, the common estimate will be a compromise between these two different parameter vectors. The corresponding score contributions will point in opposite directions, and this phenomenon can be observed through a number of different diagnostic tests, depending on the grouping of the score contributions:

- tests for pooling: in this situation groups are formed for different individuals.
- tests for structural breaks: in this situation choice pairs (a, b), with $\tau_a < \tau_b \leq \tau_0$ from early in the sample prior to a specified break date τ_0, exhibit discrepancies when compared to late pairs (a', b'), with $\tau_0 < \tau'_a < \tau'_b \leq \tau_T$, where τ_a denotes the time at which observation a occurred.
- tests for autocorrelation of the error term: as demonstrated by Büscher and Bauer (2024), the autocorrelation of the error term can be erroneously identified as a mixture in the alternative specific constants (ASCs). Consequently, discrepancies in the autocorrelation can be detected by comparing pairs with close by decisions (a, b), $|\tau_a - \tau_b| < \Delta\tau_c$ with pairs exhibiting a considerable temporal interval between them (a', b'), $|\tau_{a'} - \tau_{b'}| > \Delta\tau_d$, where $\Delta\tau_c$ and $\Delta\tau_d$ specify which temporal distances are considered to be close or distant, respectively.

Under the null hypothesis of the restricted model, the score contributions are drawn from the same distribution in all cases. Since tests for pooling at the individual level are also a standard and readily applicable procedure in the maximum likelihood setting, the subsequent analysis will focus on tests comparing different pairs of choice occasions within each individual (for which in the maximum likelihood setting, a separate model needs to be estimated for each alternative hypothesis involved in different groupings).

For the considered groups G_1 and G_2 of observation pairs, the following assumptions are made.

Assumption 3 (*Groups*)**.** The groups of pairs of observations G_1 and G_2 are constructed in such a way that:

(I) The groups G_1 and G_2 are disjoint ($G_1 \cap G_2 = \emptyset$), such that they do not contain the same pair of observations (a, b).

(II) The groups G_1 and G_2 are chosen independent of the observations y and the covariate values X_n, such that the matrix $[X_{n,a}, X_{n,b}]$ is distributed identically over all pairs (a, b) within both groups, ensuring that the model is identified locally within the respective groups (e.g. there is no perfect colinearity between regressors within groups, for example due to constant covariate values within a group when ASCs are also estimated).

(III) The groups G_1 and G_2 are chosen such that the sum of the CML weights of the pairs in group i for individual n is nonzero, i.e. $|G_{i,n}| = \sum_{a,b:(a,b)\in G_i} w_{n,a,b} > 0, i \in \{1, 2\}$.

Remark 1. It is possible for pairs with a shared observation to be contained within different groups. For instance, the pair $(1, 2)$ could be contained in the first group of subsequent observations, while the pair $(1, T)$ would belong to the group of distant observations.

Remark 2. If Assumption 3(III) is not satisfied, it is possible to remove individuals n for which $|G_{i,n}| = 0$ for some i from the subsequent calculations and adjust N in order to represent the number of individuals for which $|G_{i,n}| > 0, i \in \{1, 2\}$.

Remark 3. The weights $w_{a,b}$ used in the estimation process may impose constraints on the possible groups. To illustrate this point, consider the scenario where the weights $w_{a,b} = 0$, $|a - b| > 1$ are employed, corresponding to solely using adjacent pairs. In this instance, it is not possible to test for autocorrelation of lag greater than 1 when relying on the gradient contributions of pairs being calculated within the estimation phase. However, in this instance, the gradient contributions of all pairs can be obtained from a single evaluation of the full pairwise CML. The numerical load of such a procedure is typically small in comparison to the estimation of the model under the alternative, wherein the pairwise CML needs to be evaluated numerous times.

In order to test the null hypothesis that there are no significant differences in the expected value of the gradient contribution between two disjoint groups of CML pairs G_1 and G_2 ($G_1 \cap G_2 = \emptyset$), so

$$H_0 : \quad \mathbb{E}(g_{n,a,b}(\theta_\circ) \mid (a,b) \in G_1) = \mathbb{E}(g_{n',a',b'}(\theta_\circ) \mid (a',b') \in G_2), \tag{21}$$

one can calculate the average gradient contribution of each of the groups for decision-maker n, as well as the difference between these:

$$\bar{g}_{G_i,n} = |G_{i,n}|^{-1} \sum_{a,b:(a,b)\in G_i} w_{n,a,b}\ g_{n,a,b}(\hat{\theta}), \tag{22}$$

$$\Delta\bar{g}_{G_1,G_2,n} = \bar{g}_{G_1,n} - \bar{g}_{G_2,n}. \tag{23}$$

From Theorem 1, we have

$$\begin{aligned}\bar{g}_{G_i,n} &= |G_{i,n}|^{-1} \sum_{a,b:(a,b)\in G_i} w_{n,a,b} g_{n,a,b}(\theta_\circ) + |G_{i,n}|^{-1} \sum_{a,b:(a,b)\in G_i} w_{n,a,b} H_g(\hat{\theta} - \theta_\circ) + o_P(N^{-1/2}) \\ &= \bar{g}_{G_i,n,\circ} + H_g(\hat{\theta} - \theta_\circ) + o_P(N^{-1/2}),\end{aligned} \tag{24}$$

where the latter equation defines $\bar{g}_{G_i,n,\circ}$. It thus follows that the score contribution of the group of pairs is equal to a term that, under the null hypothesis, has zero expectation and is iid over decision-makers, plus a term that does not depend on n and is of order $O_p(N^{-1/2})$, plus a negligible term of order $o_P(N^{-1/2})$. Consequently, the difference between two groups, $\Delta\bar{g}_{G_1,G_2,n}$, is essentially equal to $\bar{g}_{G_1,n,\circ} - \bar{g}_{G_2,n,\circ}$. By taking differences, the joint term cancels under the null hypothesis.

Consequently, under the null hypothesis, a sample $\Delta\bar{g}_{G_1,G_2,n}$, $n = 1, \dots, N$ of iid observations with expectation zero is obtained. In contrast, under the alternative, the expectation will be nonzero.

Remark 4. In the aforementioned evaluations, a balanced panel situation was considered, such that the same pairs of choice occasions (a, b) were observed for each decision-maker. The weights can be specific to individual decision-makers and pairs of choices, however, provided that Assumption 2 is fulfilled. This makes the results generally applicable to unbalanced panel data and unequally weighted observations.

Although it is assumed that the pairs are interchangeable, it is challenging to quantify the distribution of the average over different pairs of decisions involved in the formation of $\bar{g}_{G_i,n}$ through analytical means. To illustrate, under the null hypothesis, the correlation between $g_{n,a,b}(\theta_\circ)$ and $g_{n,a',b'}(\theta_\circ)$ may be nonzero when $a = a'$ and zero for $a \neq a'$. Consequently, instead of utilising an estimate of the variance of $\Delta\bar{g}_{G_1,G_2,n}$ based on analytic expressions, we propose to use standard one-sample t-tests (if a single component of the gradient vector is being evaluated) and F-tests (if multiple components are assessed) for the mean, with the variance being estimated using the sample variance of the N observations. The tests are thus applicable primarily in situations where N is large.

Therefore, we propose to use the following test statistic:

Definition 1. The test statistic LM_P is defined as follows:

$$\Delta\bar{g}_{G_1,G_2} = N^{-1} \sum_{n=1}^{N} \Delta\bar{g}_{G_1,G_2,n}, \tag{25}$$

$$\hat{V}_{\Delta g} = (N-1)^{-1} \sum_{n=1}^{N} (\Delta\bar{g}_{G_1,G_2,n} - \Delta\bar{g}_{G_1,G_2})(\Delta\bar{g}_{G_1,G_2,n} - \Delta\bar{g}_{G_1,G_2})', \tag{26}$$

$$LM_P = N \Delta\bar{g}'_{G_1,G_2} \hat{V}^{-1}_{\Delta g} \Delta\bar{g}_{G_1,G_2}, \tag{27}$$

where $\Delta\bar{g}_{G_1,G_2} \in \mathbb{R}^P$.

For testing the jth parameter individually, we define the analogue of the t-test statistic:

$$t = \frac{\sqrt{N}\left(\Delta\bar{g}_{G_1,G_2,n}\right)_{(j)}}{\sqrt{(\hat{V}_{\Delta g})_{(j,j)}}}, \tag{28}$$

where $(\Delta\bar{g}_{G_1,G_2,n})_{(j)}$ denotes the jth coordinate of the vector, and $(\hat{V}_{\Delta g})_{(j,j)}$ the jth diagonal element.

With these definitions we obtain:

Theorem 2. *Under the assumptions of Theorem 1 and Assumption 3, the test statistic LM_P has an asymptotic χ^2_P distribution, given that $\hat{V}_{\Delta g}$ is regular.*

The distribution of the test statistic t converges to a standard normal distribution as $N \to \infty$, provided that $(\hat{V}_{\Delta g})_{(j,j)} \neq 0$.

For the proof of the theorem, see Appendix A.

Remark 5. It should be noted that the test statistics can be readily calculated for a number of different groupings, as the input is primarily comprised of the score contributions for each decision-maker for each pair of choices. These data are typically already calculated in gradient-based numerical optimisation procedures. Therefore, the additional computational burden associated with calculating the test statistics entails forming averages over pairs, estimating the sample mean and variance for a sample of size N, and inverting a $P \times P$ matrix.

Remark 6. To enhance the numerical stability of the calculations, it is recommended to standardise $\Delta\bar{g}_{G_1,G_2,n}$ and $\Delta\bar{g}_{G_1,G_2}$ by dividing each component by the corresponding standard deviation of $\Delta\bar{g}_{G_1,G_2,n}$ across individuals n prior to calculating $\hat{V}_{\Delta\bar{g}}$ and the test statistics. This has no impact on the test statistics; however, it does render the computation of $\hat{V}_{\Delta\bar{g}}^{-1}$ more numerically stable.

Remark 7. In case that $\hat{V}_{\Delta g}$ is singular, it is not possible to calculate the test statistics. The fact that $\hat{V}_{\Delta g}$ is singular is of intrinsic value, as it indicates that the two groups are not structurally distinct but, on the contrary, may be structurally linked. This in itself calls for further investigation by the researcher, as would the rejection of the null hypothesis by the proposed test.

A possible (extreme) example is when observations originate from a periodically repeated plan, unknown to the researcher, which results in observations from two different panel waves being identical. If groups are selected based on the panel waves, the model would be locally identified within each group, but the average score differences between the groups would be zero for all individuals, leading to $\hat{V}_{\Delta g} = 0$.

Remark 8. Under the assumptions that the $\Delta\bar{g}_{G_1,G_2,n}$ are iid multivariate normally distributed, it holds for finite samples

$$\frac{N-P}{P(N-1)} LM_P \sim F_{P,N-P}, \tag{29}$$

$$t \sim t_{N-1}. \tag{30}$$

For details, see Hotelling (1931). Asymptotically, the use of these distributions is equivalent to those in Theorem 2. However, despite the fact that the assumption of normally distributed $\Delta\bar{g}_{G_1,G_2,n}$ does not necessarily hold, the statistics in Eqs. (29) and (30) have been shown to be favourable in the simulation studies presented in Section 3.

Remark 9. In this paper, all CML estimations were performed considering all possible CML pairs (full pairwise), which were then equally weighted with $w_{n,a,b} = 1$. The number of available pairs (a, b) grows quadratically with the number of observations per decision-maker T, and consequently, the required computations per CML evaluation grow likewise when using the full pairwise approach. An overview of the required estimation times, depending on different numbers of decision-makers N and observations per decision-maker T are summarised in Table B.7 in the appendix.

A plethora of studies have investigated the effects of different weighting strategies on the computational and statistical properties of CML estimation (see, for example, Cox and Reid, 2004; Varin and Vidoni, 2005; Joe and Lee, 2009; Lindsay et al., 2011; Varin et al., 2011; Bhat, 2014; Pedeli and Varin, 2020; Büscher and Bauer, 2024). However, these effects will not be discussed in this paper, as the choice of CML pair structure is a decision taken during the model estimation stage, not the testing stage, since the computation times required to perform the Lagrange multiplier type tests proposed in this paper are negligible when compared to the estimation times (see Tables B.7 and B.8 in the appendix).

It is important to note, however, that the availability of CML pairs of differing types is required to define the different pair groups. When pairs are systematically removed from the CML estimation, this may limit the flexibility of the tests proposed in this paper. In instances where the chosen pair structure is independent of the tested CML pair groups, a reduction of CML pairs employed in the estimation process is analogous to a reduction in the number of observations per decision-maker, which is addressed in our simulation setup.

Remark 10. Note that the proposed test utilises the empirical variance $\hat{V}_{\Delta g}$ of the average groupwise gradient differences. This approach circumvents the difficult and frequently imprecise estimation of the Hessian and the variance of the score. Both of these are necessary for Wald-type tests to obtain the sandwich-type covariance matrix and they are also required for the adaptation of the log-CML ratio tests (Varin et al., 2011).

Remark 11. The Assumption 1(II) of identically distributed pairs of covariate observations is necessary to derive the distribution of the proposed test statistics under the null hypothesis. While this assumption may be violated in a number of situations, it is nevertheless plausible in many cases. Surveys using a random selection of covariates are one important example. But also in many real-world data sets, the assumption of exchangeability seems to be plausible. It is evident that the potential sources of violations to this assumption are numerous. These include, but are not limited to, the definition of groups based on covariate values or the correlation between covariates and pair groups. Consequently, the effects on the test distribution and the possible adjustments to the test are also numerous. This goes beyond the current scope of this paper and this aspect will be addressed in forthcoming publications.

3. Simulation evaluation of test properties

In order to evaluate the distributional properties of the statistics $\frac{N-P}{P(N-1)} LM_P$ and t in a finite sample setting, we employ the discrete choice modelling setting, specifically MNP models, as described in Section 1. This enables us to generate synthetic data that adheres to Assumption 1, thus allowing us to evaluate the finite sample properties of the test statistics in a controlled environment.

The synthetic data sets have been constructed in such a way as to emulate panel data sets comprising two waves of observations, with a one-year interval between the start of each wave. Specifically, when an individual has a total of T_n observations, the first set of $T_n/2$ observations starts at $\tau_1 = 1$, while the second set of $T_n/2$ observations begins at $\tau_{T_n/2+1} = 366$. Within each wave of observations, unit time steps are employed, such that $\tau_{t+1} = \tau_t + 1$.

To evaluate the properties of the statistical test devised, balanced panel data sets ($T_n = T$) with different numbers of observations per individual $T \in \{10, 14, 20, 30\}$ and different numbers of individuals $N \in \{100, 500, 1\,000\}$ were simulated.

To test the null hypothesis (21), two different time-dependent groupings of the CML pairs were considered[1]:

$$\begin{aligned} FirstLast: \quad & (a,b) \in G_1 \iff (|\tau_a| < 366) \wedge (|\tau_b| < 366) \\ & (a,b) \in G_2 \iff (|\tau_a| \geq 366) \wedge (|\tau_b| \geq 366) \\ NearFar: \quad & (a,b) \in G_1 \iff |\tau_a - \tau_b| < 100 \\ & (a,b) \in G_2 \iff |\tau_a - \tau_b| \geq 100 \end{aligned}$$

Should evidence for a difference between the gradient contributions of group G_1 and group G_2 be detected in the *FirstLast* groupings, this would indicate a structural change in the parameters of the DGP between panel waves. When considering *NearFar* groupings, a difference between the gradient contributions from the two groups could be indicative of distance-dependent correlation patterns between observations, for example induced by either structural changes between panel waves or an autoregressive error structure.

3.1. Finite sample distribution of test statistic under the null hypothesis

In order to evaluate the distribution of the test statistic under the null hypothesis, the model

$$U_{n,t,1} = 0 + \varepsilon_{n,t,1}, \tag{31}$$

$$U_{n,t,2} = \beta_0 + x_{n,t}\beta_1 + \gamma_n + \varepsilon_{n,t,2}, \tag{32}$$

was employed, with $\varepsilon_{n,t} = (\varepsilon_{n,t,1}, \varepsilon_{n,t,2})' \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & \sigma_\varepsilon^2 \end{pmatrix}\right)$ and $\gamma_n \sim \mathcal{N}(0, \sigma_\gamma^2)$. For the DGP we set $\beta_0 = 1$, $\beta_1 = 1$, $\sigma_\gamma = 1$, and $\sigma_\varepsilon = 1$. In order to ensure model identification, we kept $\beta_1 = 1$ fixed during estimation and estimated $\theta = (\beta_0, \sigma_\gamma, \sigma_\varepsilon)$. The regressors $x_{n,t}$ were drawn iid from a standard normal distribution.

3.2. Detecting autoregressive errors

In this scenario, the DGP is specified as

$$U_{n,t,1} = 0 + \varepsilon_{n,t,1}, \tag{33}$$

$$U_{n,t,2} = \beta_0 + x_{n,t}\beta_1 + \varepsilon_{n,t,2}, \tag{34}$$

$$\varepsilon_{n,t} = (\varepsilon_{n,t,1}, \varepsilon_{n,t,2})' = \rho\varepsilon_{n,t-1} + \tilde{\varepsilon}_{n,t}, \tag{35}$$

with $\tilde{\varepsilon}_{n,t} = (\tilde{\varepsilon}_{n,t,1}, \tilde{\varepsilon}_{n,t,2})' \sim \mathcal{N}\left(0, \begin{pmatrix} 1-\rho^2 & 0 \\ 0 & 1-\rho^2 \end{pmatrix}\right)$, $\beta_0 = 1$, $\beta_1 = 1$, and varying values of ρ between -0.99 and 0.99 for different simulation setups.[2] The regressors $x_{n,t}$ are drawn iid from a standard normal distribution.

The estimation is then based on the misspecified model

$$U_{n,t,1} = 0 + \gamma_{n,1} + \varepsilon_{n,t,1}, \tag{36}$$

$$U_{n,t,2} = \beta_0 + x_{n,t}\beta_1 + \gamma_{n,2} + \varepsilon_{n,t,2}, \tag{37}$$

with $\varepsilon_{n,t} = (\varepsilon_{n,t,1}, \varepsilon_{n,t,2})' \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & \sigma_\varepsilon^2 \end{pmatrix}\right)$ and $\gamma_n \sim \mathcal{N}(0, \sigma_\gamma^2 I_2)$, with $\beta_1 = 1$ fixed for identification and estimated $\theta = (\beta_0, \sigma_\gamma, \sigma_\varepsilon)$.

[1] Additionally, a third type of grouping, termed *EvenOddClose*, was explored. This involved classifying pairs of observations with even temporal distances from the same panel wave in one group and pairs of observations with odd temporal distances between them from the same panel wave in a second group. These, however, did not demonstrate any advantages compared to the other groupings described in this paper, in any of the scenarios considered in the finite sample simulation studies. Further details can be provided upon request.

[2] Values used for ρ were $\{-0.990, -0.950, -0.900, -0.750, -0.625, -0.500, -0.400, -0.300, -0.200, -0.100, -0.050, -0.025, -0.010, 0.000, 0.010, 0.025, 0.050, 0.100, 0.200, 0.300, 0.400, 0.500, 0.625, 0.750, 0.900, 0.950, 0.990\}$.

The covariance matrix $\Sigma_{n,a,b} = \mathrm{Cov}(\varepsilon_{n,a}, \varepsilon_{n,b})$ between the random errors of two observations from one individual can now be explicitly expressed for both models. In general, for an autoregressive error process $\varepsilon_{n,t} = \Psi\varepsilon_{n,t-1} + \tilde{\varepsilon}_{n,t}$ for equidistant time steps with $\tau_t = \tau_{t-1} + 1$ and $\tilde{\varepsilon}_{n,t} \sim \mathcal{N}(0, \tilde{\Sigma})$, we have

$$\Sigma_{n,a,b} = \begin{pmatrix} \Sigma & \Sigma(\Psi^{\tau_b-\tau_a})' \\ \Psi^{\tau_b-\tau_a}\Sigma & \Sigma \end{pmatrix}, \tag{38}$$

where Σ solves the Lyapunov equation $\Sigma = \Psi\Sigma\Psi' + \tilde{\Sigma}$, provided that all eigenvalues of the matrix $\Psi \in \mathbb{R}^{J\times J}$ are inside the unit circle (stability condition).

For the DGP model, we have $\Psi = \rho I_J$ and $\tilde{\Sigma} = (1-\rho^2)I_J$, where I_J represents the $J-$dimensional identity matrix. Consequently,

$$\Sigma_{n,a,b} = \begin{pmatrix} \frac{1}{1-\rho^2}\tilde{\Sigma} & \frac{\rho^{|\tau_b-\tau_a|}}{1-\rho^2}\tilde{\Sigma} \\ \frac{\rho^{|\tau_b-\tau_a|}}{1-\rho^2}\tilde{\Sigma} & \frac{1}{1-\rho^2}\tilde{\Sigma} \end{pmatrix} = \begin{pmatrix} 1 & 0 & \rho^{|\tau_b-\tau_a|} & 0 \\ 0 & 1 & 0 & \rho^{|\tau_b-\tau_a|} \\ \rho^{|\tau_b-\tau_a|} & 0 & 1 & 0 \\ 0 & \rho^{|\tau_b-\tau_a|} & 0 & 1 \end{pmatrix}. \tag{39}$$

In contrast, the estimated model yields the following results:

$$\Sigma_{n,a,b} = \begin{pmatrix} 1+\sigma_\gamma^2 & 0 & \sigma_\gamma^2 & 0 \\ 0 & \sigma_\varepsilon^2+\sigma_\gamma^2 & 0 & \sigma_\gamma^2 \\ \sigma_\gamma^2 & 0 & 1+\sigma_\gamma^2 & 0 \\ 0 & \sigma_\gamma^2 & 0 & \sigma_\varepsilon^2+\sigma_\gamma^2 \end{pmatrix} \tag{40}$$

and

$$\mathrm{Cor}(\varepsilon_{n,a}, \varepsilon_{n,b}) = \begin{pmatrix} 1 & 0 & \sigma_\gamma^2/(1+\sigma_\gamma^2) & 0 \\ 0 & 1 & 0 & \sigma_\gamma^2/(\sigma_\varepsilon^2+\sigma_\gamma^2) \\ \sigma_\gamma^2/(1+\sigma_\gamma^2) & 0 & 1 & 0 \\ 0 & \sigma_\gamma^2/(\sigma_\varepsilon^2+\sigma_\gamma^2) & 0 & 1 \end{pmatrix}. \tag{41}$$

Under the null hypothesis of an exchangeable probit model, ρ is equal to zero, thereby rendering the two models identical for $\sigma_\gamma = 0$. When $\rho \neq 0$, a notable distinction between the two models emerges: the estimated model assumes a constant covariance matrix for all pairs of observations, irrespective of the temporal distance between them. In contrast, the DGP models the covariance depends on the distance $|\tau_b - \tau_a|$. When observations a and b are in different panel waves, we have $|\rho^{|\tau_b-\tau_a|}| \leq 0.99^{366-30} \approx 0.0342$, whilst for $\rho = 0.99$ we have for observations within the same panel wave, $0.99^{|\tau_b-\tau_a|} \geq 0.99^{|30-1|} \approx 0.7472$.

Furthermore, the estimated model is not sufficiently flexible to estimate negative correlations between errors at different time points. However, this is the case for pairs of observations with $|\tau_b - \tau_a| \equiv 1 \pmod 2$ for negative values of ρ.

3.3. Detecting structural breaks in the model

In this scenario, the DGP under consideration is

$$U_{n,t,1} = 0 + \varepsilon_{n,t,1}, \tag{42}$$

$$U_{n,t,2} = \beta_0 + x_{n,t}\beta_{1,n,t} + \gamma_n + \varepsilon_{n,t,2}, \tag{43}$$

with $\varepsilon_{n,t} = (\varepsilon_{n,t,1}, \varepsilon_{n,t,2})' \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & \sigma_\varepsilon^2 \end{pmatrix}\right)$ and $\gamma_n \sim \mathcal{N}(0, \sigma_\gamma^2)$. For the DGP, the values $\beta_0 = 1$, $\sigma_\gamma = 1$, and $\sigma_\varepsilon = 1$ were used. In order to examine the effects of different groupings of individuals with varying parameter values between the groups and changes in these parameter values over time, we implemented a series of scenarios involving $\beta_{1,n,t}$. This temporal change was chosen to represent a discrete change in the parameter values between the two simulated panel waves. Two distinct configurations were employed.

The first configuration represents an overall shift of the parameter value for the entire population between the panel waves, so

$$\beta_{1,n,t} = \begin{cases} 1-\alpha, & \tau_t < 365, \\ 1+\alpha & \tau_t \geq 365. \end{cases} \tag{44}$$

The second setup represents a split of the population. The population is partitioned into two equally sized groups P_1 and P_2. In the initial panel wave, the parameter value was identical for all individuals. In contrast, in the second panel wave, the parameter value was increased for those individuals in P_1 and decreased for those in P_2, resulting in

$$\beta_{1,n,t} = \begin{cases} 1, & \tau_t < 365, \\ 1+\alpha, & \tau_t \geq 365 \text{ and } n \in P_1, \\ 1-\alpha & \tau_t \geq 365 \text{ and } n \in P_2. \end{cases} \tag{45}$$

In order to evaluate the rejection rate of the proposed test for different magnitudes of violations of the null hypothesis of an exchangeable probit model ($H_0 : \alpha = 0$), data sets with varying values of α were simulated.[3]

For estimation, the misspecified model

$$U_{n,t,1} = 0 + \varepsilon_{n,t,1}, \tag{46}$$

$$U_{n,t,2} = \beta_0 + x_{n,t}\beta_1 + \gamma_n + \varepsilon_{n,t,2}, \tag{47}$$

was employed, with $\varepsilon_{n,t} = (\varepsilon_{n,t,1}, \varepsilon_{n,t,2})' \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & \sigma_\varepsilon^2 \end{pmatrix}\right)$ and $\gamma_n \sim \mathcal{N}(0, \sigma_\gamma^2)$. The value of $\sigma_\varepsilon = 1$ was fixed for identification purposes, while $\theta = (\beta_0, \beta_1, \sigma_\gamma)$ was estimated. Here, a common value of $\beta_{1,n,t} = \beta_1$ was assumed for all individuals and time points.

4. Simulation results

This section presents a discussion of the results obtained from the simulation studies described in Section 3. The results are presented in tabular form in Appendix B and illustrated in the accompanying plots.

4.1. Distribution under the null hypothesis

In the plots in Fig. B.1, the empirical CDFs of the test statistics are compared to their theoretical CDFs, as deduced in Section 2. Each figure represents a single test statistic. The Fig. B.1(a) for the joint test statistic is based on the specification in Eq. (29), while the three Figs. B.1(b)–B.1(d) for the componentwise tests are based on Eq. (28). Each plot contains a matrix of 12 plots, with one plot for each simulated combination of the number of individuals $N \in \{100, 500, 1\,000\}$ and the number of observations per individual $T \in \{10, 14, 20, 30\}$. The different groupings that were considered are represented in the plots by differently coloured solid lines, whilst the theoretical distribution is overlain as a black dashed line. The x-axis represents the value of the test statistic, and the y-axis represents the corresponding CDF value. The plots demonstrate that for all 12 considered combinations of N and T, the empirical distribution of the test statistic closely follows its theoretical counterpart, irrespective of the number of individuals and the number of observations per individual.

Kolmogorov–Smirnov tests (for details, see Kolmogorov, 1933; Marsaglia et al., 2003) were performed for each of the empirical distributions against their theoretical asymptotic counterparts, resulting in test statistics in the range of 0.016 to 0.061 with a critical value at a 5% significance level of $K_{0.05} = 1.3581/\sqrt{N} = 0.043$. To address the issue of multiple testing, a Bonferroni correction was applied (Bonferroni, 1936). This revealed that the null hypothesis, namely that the simulated test statistics are realisations of the theoretical asymptotic test distributions, was not rejected for any of the cases.

4.2. Rejections rates under violation of the null hypothesis

Figs. B.2–B.4 illustrate the rejection rates of the tests at a significance level of 5% on the y-axis, with the degree of violation of the null hypothesis on the x-axis (autocorrelation coefficient ρ for the autoregressive error case and the shift/split value α for the simulations with shifts/splits between panel waves). Each of the figures represents one of the simulated violations of the null hypothesis. These are as follows: B.2 for the autoregressive error process, B.3 for shifts in a parameter between panel waves, and B.4 for a split of the population between panel waves, with both halves of the population having a unique parameter value.

Figs. B.2(a)–B.2(b), B.3(a)–B.3(b), and B.4(a)–B.4(b) contain a grid of twelve graphs arranged by number of individuals N and number of observations T per individual. Each point in the plots is based on tests on 100 simulated data sets, except for the cases with $N = 500$ and $T = 20$, where 1 000 simulated data sets were used for each point. Figs. B.2(c)–B.2(d), B.3(c)–B.3(d), and B.4(c)–B.4(d) are larger-sized plots for the aforementioned cases with $N = 500$ and $T = 20$, which also contain data for a greater number of values of ρ and α, respectively. The different tests (joint and componentwise) are represented in different colours within the plots.

For the cases exhibiting a parameter shift between panel waves or a parameter split in the second panel wave, for the setup with $N = 500$ and $T = 20$, a Wald test was also computed for the *FirstLast* grouping, for each combination of violation type and severity, for 100 of the data sets to compare rejection rates and required testing times with a widely used benchmark test. As shown in Fig. B.5 of the Appendix, the rejection rates of the Wald test are for both violations comparable to those of the proposed joint LM type test. However, a comparison of the required computation times for the calculation of the tests reveals the advantages of the LM type test. The calculation of the Wald test requires the estimation of the unrestricted model, whilst the LM type test relies on gradient contributions readily available from the estimation of the restricted model. The estimation of the unrestricted model was performed by using the estimated parameter values of the restricted model as starting values and subsequently estimating distinct parameter values for the first panel wave and the second panel wave. The calculation times for the proposed LM type test, in addition to the time needed for estimating the restricted model, ranged from 0.16 s to 0.36 s, with a mean of 0.23 s. In contrast, the calculation of a Wald type test, in addition to the time needed for estimating the restricted model, required between 113.07 s and 327.91 s, with a mean of 180.35 s. On average, the LM type test was 793.98 times faster than the Wald type test in the evaluated cases (see Table B.8 for details).

[3] The values used for α were {0.0000, 0.0100, 0.0175, 0.0250, 0.0500, 0.0750, 0.1000, 0.1750, 0.2500, 0.3750, 0.5000, 0.7500, 0.9500, 1.0000, 1.5000, 2.0000}.

Additionally, for each simulation scenario and grouping of CML pairs, the componentwise test with the largest test statistic was identified. Tables B.1–B.6 provide an overview of the respective shares across the 100 (1 000 for $N = 500, T = 20$) simulated cases where the largest test statistic among the componentwise tests was observed for the given component.

4.2.1. Rejection rate with autoregressive errors

For the cases with an autoregressive error structure, the tests with the *FirstLast* groupings do not exhibit rejection rates that are substantially larger than 5% for any values of ρ, as illustrated in Figs. B.2(a) and B.2(c). This is to be expected, given that there are no structural differences between the first and second waves of observations. This is also reflected in the fact that no componentwise test yielded a clear majority of shares with the largest test statistics, as can be seen in Table B.1.

The tests with the *NearFar* groupings demonstrate that, for both the joint test and the test for the parameter σ_γ (the variance parameter for the mixed ASC effect), the rejection rates increase with the absolute value of the autocorrelation coefficient ρ. As the number of individuals N increases, this effect becomes more pronounced. However, an increase in the number of observations, coupled with an extreme negative autocorrelation, results in a decline in the rejection rate, as illustrated in Figs. B.2(b) and B.2(d). This can be attributed to the fact that with an increased number of observations per decision-maker, the maximum distance between observations within a group increases, and the average correlation between observations within these pairs already diminishes. Furthermore, the models specified for estimation are not capable of accounting for negative correlation between error terms of different observations.

The tests for β_0 and σ_ϵ fail to detect violations of the null hypothesis for negative autocorrelation coefficients ρ. However, they do exhibit an increase in rejection rates with larger positive autocorrelations.

The results presented in Table B.2 also demonstrate that the test for σ_γ, which represents the variance parameter for the mixed effects, predominantly yields the largest test statistic. Notably, the share of cases where σ_γ has the largest test statistic approaches one as the absolute value of the autocorrelation coefficient ρ increases. This suggests to the practitioner that when the largest componentwise test statistic is connected to the correlation between observations from the same individual, the violation of the null hypothesis is also connected to how the correlation between observations is implemented in the model.

The sole exceptions are cases with extreme negative autocorrelation ($\rho \leq -0.95$), where the share of cases where the test for β_0 had the largest test statistic does rise with rising number of observations per individual T.

4.2.2. Rejection rate with shift of parameter value between panel waves

For the cases where there is a shift in the β_1 parameter between panel waves (Fig. B.3), the tests with the *FirstLast* groupings demonstrate the strongest increase in rejection rates for an increasing degree of violation of the null hypothesis, reaching a rejection rate of 1 for $\alpha \geq 0.5$ across all tests and combinations of N and T. The only exception is the test for the ASC parameter β_0, which does not achieve a perfect rejection rate. As the number of individuals N or the number of observations per individual T increases, the perfect rejection rate is reached for even smaller values of α, as illustrated in Figs. B.3(a) and B.3(c). The share of cases in which the test for the β_1 parameter exhibits the largest test statistic increases rapidly to one as the magnitude α of the shift in the parameter β_1 between panel waves is increased. This effect is more pronounced in larger sample sizes (larger values of N or T).

The results obtained with the *NearFar* groupings are less pronounced, yet still demonstrate a clear and conclusive pattern. For all tests, the rejection rates increase with rising values of α and with larger numbers of individuals N and observations per individual T. Across the board, the joint test shows the highest rejection rates, followed by the test for the variance parameter of the mixed ASC σ_γ (see Figs. B.3(b) and B.3(d)). This is consistent with expectations, as pairs of observations within the same panel wave are expected to exhibit higher correlation and, consequently, higher values of σ_γ than pairs of observations in different panel waves. This is due to the fact that the β_1 parameter changes values between two panel waves for all individuals. The effect is less pronounced for tests of the parameters of β_0 and β_1, and the dependence of high rejection rates on larger numbers of individuals N and observations per individual T is stronger. For the *NearFar* groupings, the share of cases in which the test for the σ_γ parameter exhibits the largest test statistic increases when the magnitude α of the shift in the parameter β_1 between panel waves is increased, reaching a value of one for large data sets. This is in contrast to the results obtained with the *FirstLast* groupings, where the test for β_1 had the largest shares.

The collective findings assist the practitioner in comprehending the underlying cause of the violation of the null hypothesis. The evidence suggests that pairs of observations from different panel waves exhibit different β_1 values, which, in turn, would explain the observed variation in correlations between observations with varying temporal distances, attributable to the differing σ_γ parameters. Observation pairs from the same panel display greater similarity in their β_1 parameters, leading to higher levels of correlation compared to observation pairs from different panel waves, which exhibit greater variation between their β_1 parameters.

4.2.3. Rejection rate with split of parameter value in second panel wave

The cases with a split in the β_1 parameter in the second panel wave (Fig. B.4) demonstrate comparable outcomes for the *FirstLast* grouping to those observed for the rejection rates with the shift violations detailed in Section 4.2.2, albeit a more pronounced degree of violation is required for the tests to reject the null hypothesis. Even for the largest combination of number of individuals ($N = 1\,000$) and number of observations per individual ($T = 30$) tested, the rejection rates did not differ significantly from 0.05 for $\alpha \leq 0.175$. The observed rejection rates depending on the degree of violation of the null hypothesis α can be roughly compared to sigmoid functions, where larger values of N and T result in inflection points for lower values of α. The joint test and the tests for β_1 and σ_γ exhibit comparable rejection rates, whereas the rejection rate of the test for the ASC parameter β_0 requires higher values of α to reliably reject the null hypothesis (see Figs. B.4(a) and B.4(c)). As demonstrated in Table B.5, the test for the β_1 parameter

exhibits a similar pattern to that observed in the case of a shift of the β_1 parameter as discussed in Section 4.2.2. Specifically, the test for the β_1 parameter has the largest share of cases with the highest test statistic when the magnitude of the violation of the null hypothesis, as measured by the value of α, increases.

The tests with the *NearFar* groupings (Figs. B.4(b) and B.4(d)) demonstrate the greatest dependency on the number of individuals and the number of observations per individual. Notably, the tests for $N = 100$ individuals exhibit no higher rejection rates than would be expected under the null hypothesis, regardless of the degree of violation α. For $N = 1\,000$ and $T = 30$, however, a rejection rate of 1 is reached with $\alpha \geq 1.0$ for the joint test. For this grouping and type of violation, our simulations did not indicate that a single parameter was dominating the test statistics, as evidenced by the results in Table B.6.

5. Conclusion

In this paper, we have constructed a Lagrange multiplier-type test for the composite marginal likelihood estimation framework, to test for intra-personal heterogeneity, derived the asymptotic test distribution under the null hypothesis, and, using synthetic data sets, verified the test distribution in finite samples and evaluated the size and power of the test under different types and severities of violations of the null hypothesis.

The theoretical derivations demonstrate how the concept of testing for pooling using LM tests can be extended to the CML estimation setting, thereby enabling testing at the pair level rather than the individual level. The proposed procedures require minimal additional computation time and permit the testing of multiple hypotheses without the necessity of estimating or even implementing the unrestricted models, thereby making them a versatile testing tool.

The simulation results demonstrate that the proposed tests follow the asymptotic test distributions under the null hypothesis already for relatively small samples closely. Furthermore, with an appropriate sample size, the violations of the null hypothesis considered in this paper can be detected reliably, even when the degree of violation is modest. In the case of smaller sample sizes (small N and/or small T), the rejection rate of the proposed tests only approaches 1 for stronger degrees of violation of the null hypothesis.

It was also demonstrated that, in the finite sample setting, rejection rates under violation of the null hypothesis are comparable with those of the established Wald type test. This is accompanied by a substantial reduction in calculation times, with an average reduction by a factor of 794 or about three minutes being observed in the cases examined.

In practice, when the type of violation that may be present in the data is typically unknown, the test can also be used to infer which violation may be responsible for a rejection of the null hypothesis. In instances where the tests utilising the *NearFar* groupings reject the null hypothesis, yet tests employing the *FirstLast* groupings do not, this may be indicative of a change in the correlation between pairs of observations with different temporal lags, for example as induced by an autoregressive error process. In instances where tests for both types of groupings, *FirstLast* and *NearFar*, reject the null hypothesis, a potential explanation for this could be a change in parameter values between panel waves, as this introduces structural differences between the two panel waves, but also reduces the correlation between observations from different panel waves. In examining which individual parameter test yielded the largest test statistic, a practitioner may gain greater insight than would be possible through a joint test. The parameter with the largest test statistic can provide information regarding the location of the largest discrepancy between the two groups of observational pairs. Depending on the chosen grouping, this can inform the practitioner of where in the model the misspecification responsible for the rejection of the null hypothesis may occur.

In accordance with standard diagnostic procedures, the subsequent steps for a practitioner would be to adjust the model according to the detected misspecification, compare the adjusted model with the original model via a model selection criterion (e.g., the composite likelihood ratio test (CLRT) or the composite likelihood Bayesian information criterion (CLBIC), see Gao and Song, 2010), and possibly accept the new model. This would then be tested in turn for misspecifications, thus starting the loop anew. Alternatively, one could employ estimation techniques that are robust against the detected type of misspecification, like the weighted CML estimator described by Büscher and Bauer (2024) for panel data with an autoregressive error structure.

Avenues for further research include the investigation of the potential application of the proposed tests in the context of seasonality assessment in discrete choice data sets. This subject matter is of particular relevance within the time series literature (compare Franses and Paap, 2004; Hyndman and Athanasopoulos, 2018, for example). A further investigation into the statistical properties of the LM tests proposed in this paper could include the analysis of the effects of different sources for the violation of the assumptions of identical distribution of the CML pair gradient contributions on the test statistics distribution and how these can be identified and mitigated. Investigating a preliminary stage of model diagnostics to be used prior to the proposed tests, comparable to residual plots in the linear model setting, can also be beneficial and can help practitioners to identify groups of observational pairs to be used in the tests proposed in this paper, as was shown by Büscher (2025), who developed gradient contribution based diagnostic plots to identify non-linearities in the utility function and dynamic and temporal effects in the data. In one example, the diagnostic plots were employed in an applied example to identify groups of observations within a data set. Subsequently, the LM tests proposed in this paper were implemented to verify significant differences between these groups. This example demonstrates that, in an applied setting, practitioners can integrate the CML Lagrange multiplier tests into their choice modelling workflow. The integration of these tests into a choice modelling workflow is further elaborated in Büscher (2024), Chapter 5.

CRediT authorship contribution statement

Sebastian Büscher: Visualization, Methodology, Conceptualization, Writing – original draft, Software, Formal analysis, Writing – review & editing, Validation, Investigation. **Dietmar Bauer:** Writing – review & editing, Project administration, Funding acquisition, Software, Methodology, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Theorem 1. *Let the data be generated from a mixed MNP model with utility* $U_{n,t,j} = X'_{n,t,j}\beta + X'_{n,t,j}\gamma_n + \epsilon_{n,t,j}$ *for* $n = 1, \dots, N$, $t = 1, \dots, T_n$, *where* $\gamma_n \sim \mathcal{N}(0, \Omega)$ *independent of* $X_{n,t,j}$, *and* $\epsilon_{n,t,j}$ *iid over decision-makers, such that Assumption 1 is satisfied.*

Let the model be estimated using CML with weights $w_{n,a,b}$ *in accordance with Assumption 2, providing the estimator* $\hat{\theta}$ *maximising the CML such that the score is zero and the estimator is asymptotically normal with* $\sqrt{N}(\hat{\theta} - \theta_\circ) \xrightarrow{d} Z \sim \mathcal{N}(0, V_\theta)$.

Then

$$g_{n,a,b}(\hat{\theta}) = \partial_\theta \log L_{n,a,b}(\hat{\theta}) = g_{n,a,b}(\theta_\circ) + \partial_\theta g_{n,a,b}(\bar{\theta}_n)(\hat{\theta} - \theta_\circ), \tag{19}$$

where $\mathbb{E}(g_{n,a,b}(\theta_\circ)) = 0$. *Furthermore,* $\mathbb{E}\partial_\theta g_{n,a,b}(\theta_\circ) = H_g > 0$ *and* $\bar{\theta}_n$ *is an intermediate value such that* $\bar{\theta}_n \xrightarrow{p} \theta_\circ$ *as* $N \to \infty$.

Finally,

$$N^{-1} \sum_{n=1}^{N} \partial_\theta g_{n,a,b}(\bar{\theta}_n)(\hat{\theta} - \theta_\circ) = H_g(\hat{\theta} - \theta_\circ) + o_P(N^{-1/2}). \tag{20}$$

Proof. The intermediate value theorem provides the approximation of $g_{n,a,b}(\hat{\theta})$. Correct specification of the model implies that the score at the true parameter vector has zero expectation and the Hessian is positive definite. The data generating process implies that the Hessian H_g is identical for all pairs.

The last statement then holds from standard asymptotic theory since $N^{-1} \sum_{n=1}^{N} \partial_\theta g_{n,a,b}(\bar{\theta}) \to H_g$ for each sequence $\bar{\theta} \to \theta_\circ$. Since the intermediate points all lie between $\hat{\theta}$ and $\theta_\circ$ the maximal distance of the intermediate points decreases to zero. □

Theorem 2. *Under the assumptions of Theorem 1 and Assumption 3, the test statistic* LM_P *has an asymptotic* χ^2_P *distribution, given that* $\hat{V}_{\Delta g}$ *is regular.*

The distribution of the test statistic t *converges to a standard normal distribution as* $N \to \infty$, *provided that* $(\hat{V}_{\Delta g})_{(j,j)} \neq 0$.

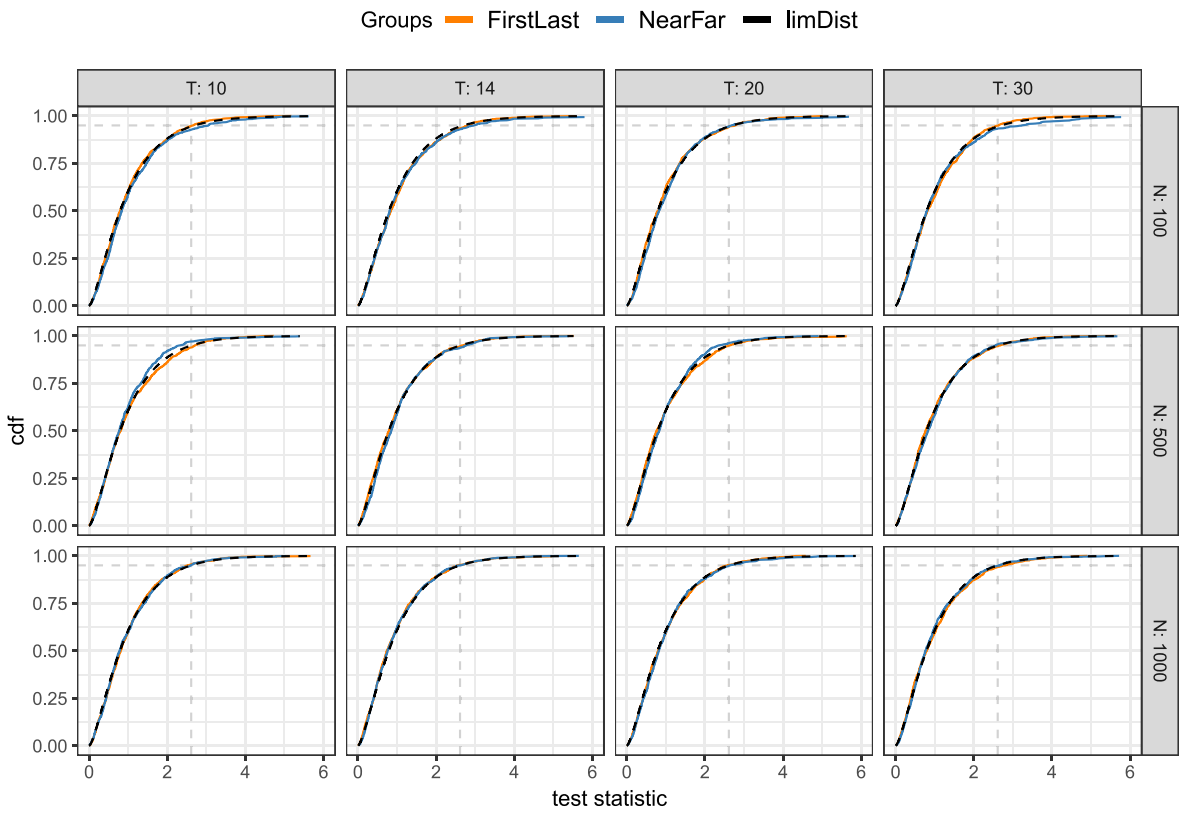
Proof. The theorem follows from the iid property of $\Delta\bar{g}_{G_1,G_2,n}$ in combination with bounds on the variances. Since the regressor vector has bounded support, the gradients are uniformly bounded, such that standard central limit theorems for iid sequences can be applied.

The consistency of the estimation of the variance follows from the law of large numbers applied to the empirical variance.

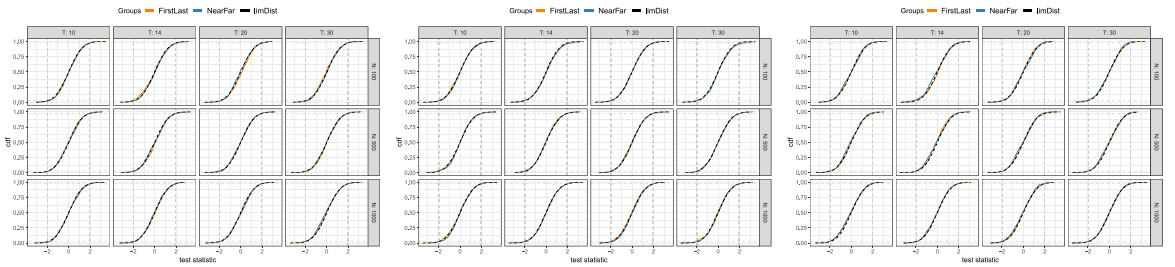
For the squared test statistic LM_P, we hence obtain asymptotic χ^2_P distribution from standard arguments. The t-statistic converges to a standard normal distribution. □

Appendix B

This appendix section contains the tables indicating the shares of the largest test statistic among the componentwise tests, the plots showcasing the distribution of the test statistics under the null hypothesis, and the plots of the rejection rates depending on the severity of violation of the null hypothesis, summarising the empirical results of the paper (see Figs. B.1–B.5 and Tables B.1–B.8).

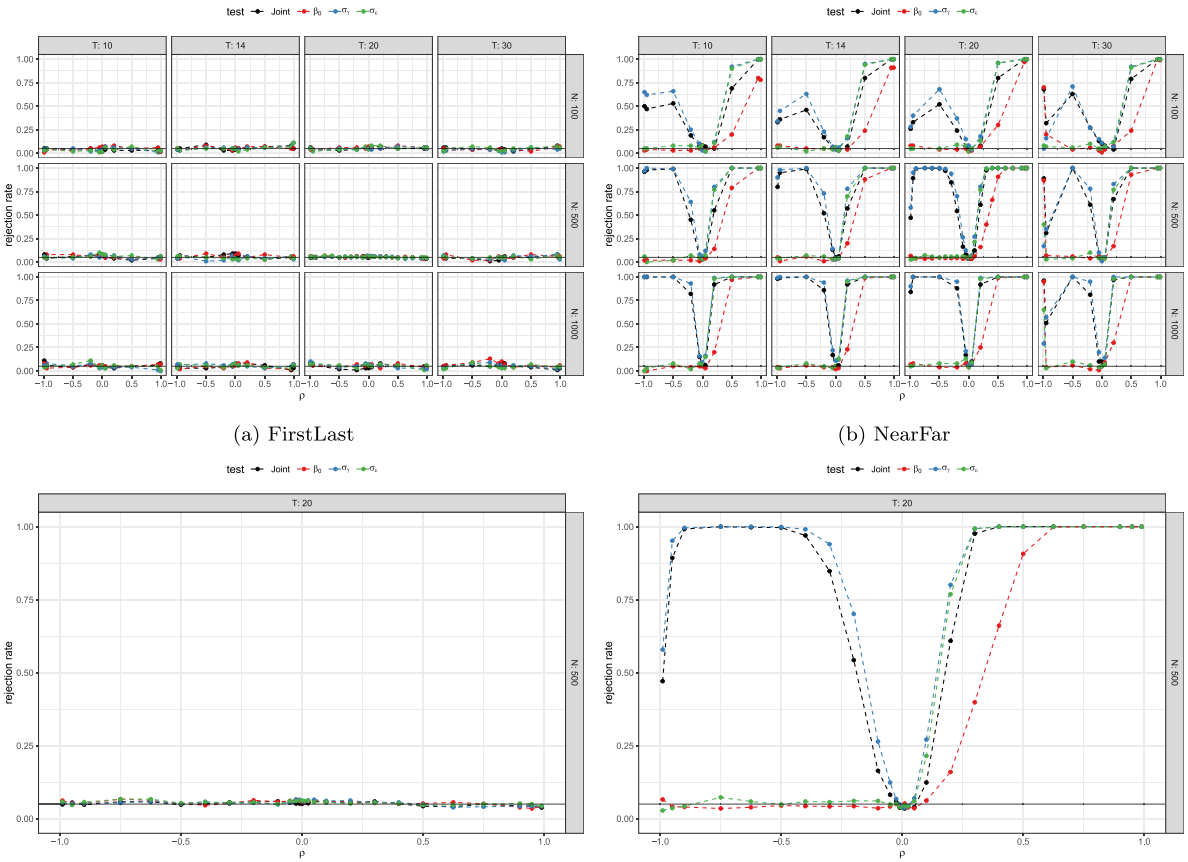


(a) Joint test



(b) $\theta_1 = \beta_0$ (c) $\theta_2 = \sigma_\gamma$ (d) $\theta_3 = \sigma_\varepsilon$

Fig. B.1. CDFs of the test statistics for the joint test and the tests for the individual parameter components of $\theta = (\beta_0, \sigma_\gamma, \sigma_\varepsilon)'$ for the three different groupings. These are based on 1 000 simulated data sets each, with the CDF of the limiting distribution of the test statistic shown in dashed black. The dashed vertical lines represent the respective 95% quantile for the joint test and the 2.5% and 97.5% quantiles for the componentwise tests of the theoretical and empirical distributions. The dashed horizontal lines represent the value of the respective empirical CDFs evaluated at the 95% quantile for the joint test and the 2.5% and 97.5% quantiles for the componentwise tests of the theoretical distribution.



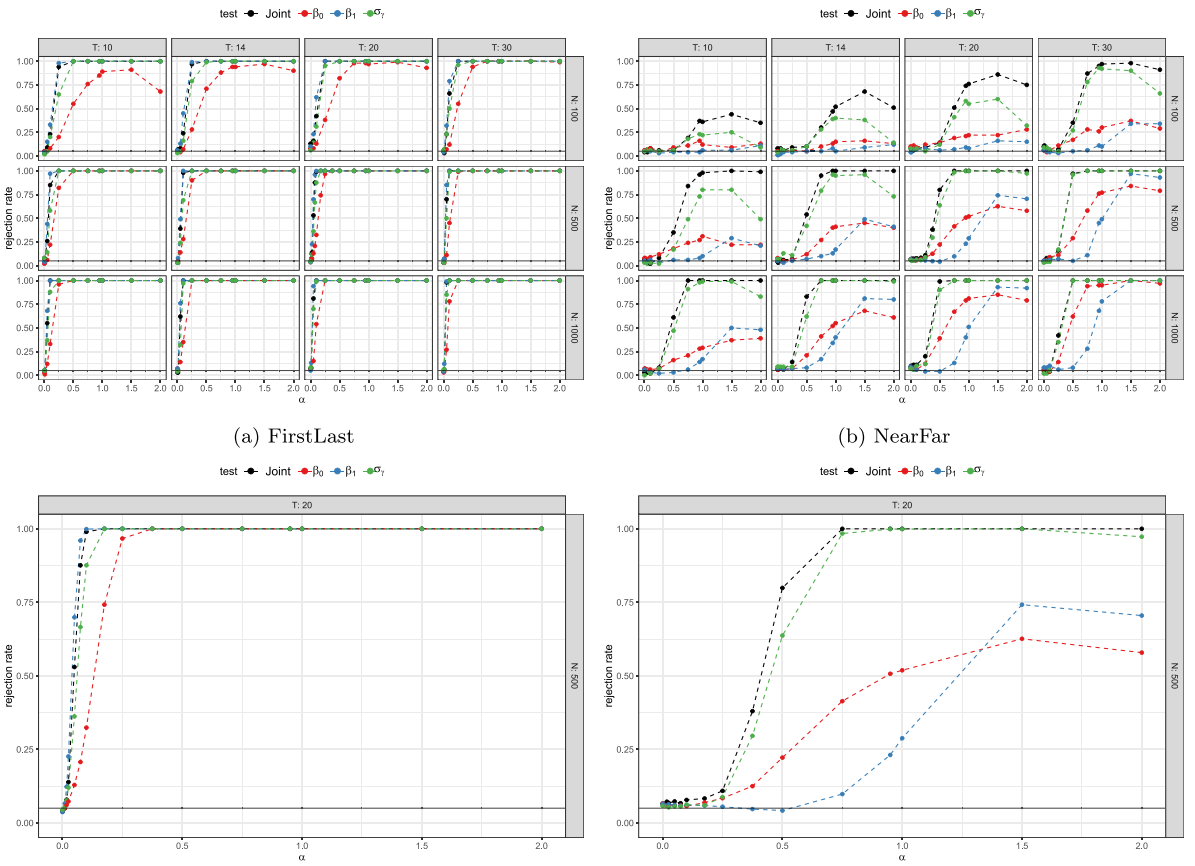
(a) FirstLast

(b) NearFar

(c) FirstLast with $T = 20$, $N = 500$

(d) NearFar with $T = 20$, $N = 500$

Fig. B.2. Rejection rates of the proposed tests for different CML-pair groupings, with the different tests separated by colour and the autocorrelation coefficient of the autoregressive errors on the x-axis. The rejection rates presented herein are based on 100 simulated data sets, with the exception of those with $T = 20$ and $N = 500$, for which 1 000 simulations were employed.



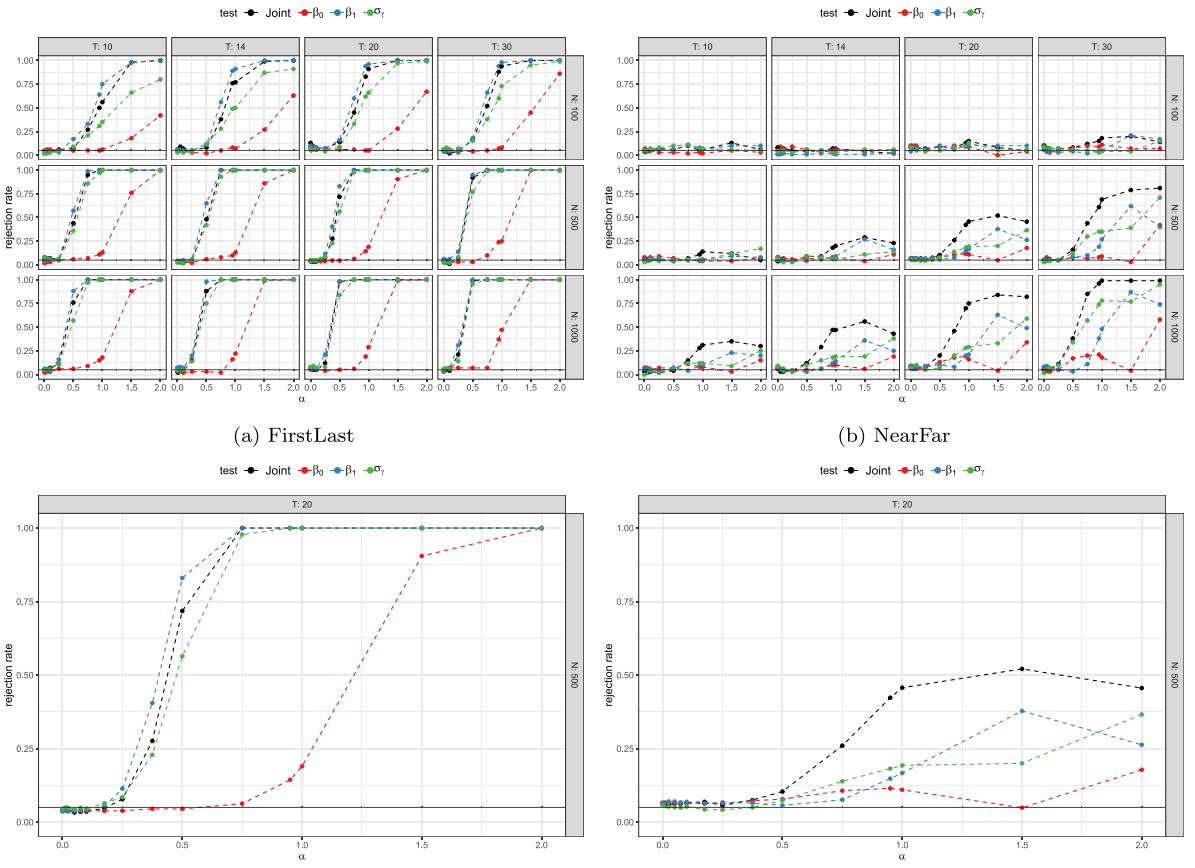
(a) FirstLast

(b) NearFar

(c) FirstLast with $T = 20$, $N = 500$

(d) NearFar with $T = 20$, $N = 500$

Fig. B.3. Rejection rates of the proposed tests for different CML pair groupings, with the different tests separated by colour and the severity of the violation of the null hypothesis (shift of $\beta_{1,n,t}$ value between panel waves) on the x-axis. The rejection rates presented herein are based on 100 simulated data sets, with the exception of those with $T = 20$ and $N = 500$, for which 1 000 simulations were employed.



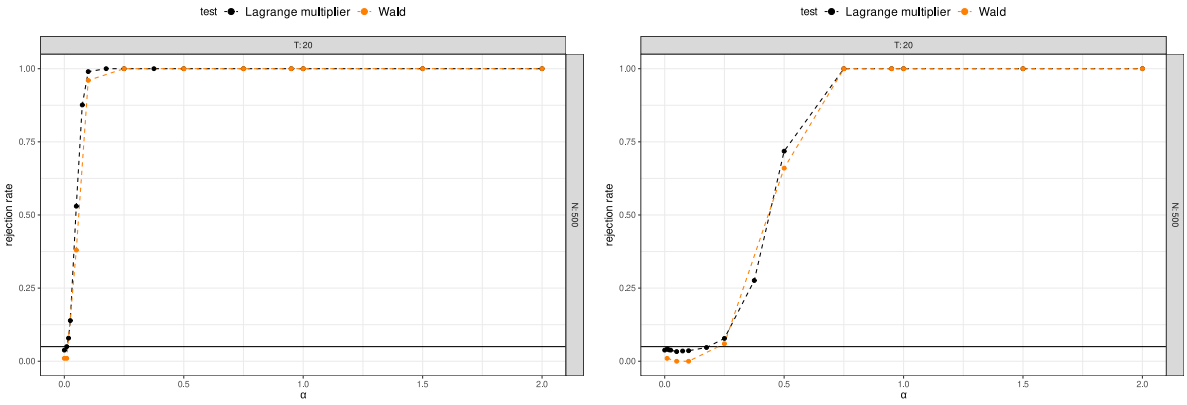
(a) FirstLast

(b) NearFar

(c) FirstLast with $T = 20$, $N = 500$

(d) NearFar with $T = 20$, $N = 500$

Fig. B.4. Rejection rates of the proposed tests for different CML pair groupings, with the different tests separated by colour and the severity of the violation of the null hypothesis (split of $\beta_{1,n,t}$ value between groups of individuals in second panel wave) on the x-axis. The rejection rates presented herein are based on 100 simulated data sets, with the exception of those with $T = 20$ and $N = 500$, for which 1 000 simulations were employed.



(a) Parameter shift, $\beta_{1,n,t} = \begin{cases} 1-\alpha, & \tau_t < 365, \\ 1+\alpha & \tau_t \geq 365. \end{cases}$

(b) Parameter split, $\beta_{1,n,t} = \begin{cases} 1, & \tau_t < 365, \\ 1+\alpha, & \tau_t \geq 365 \text{ and } n \in P_1, \\ 1-\alpha & \tau_t \geq 365 \text{ and } n \in P_2. \end{cases}$

Fig. B.5. Rejection rates of the proposed joint CML Lagrange multiplier type test and a Wald type test the *FirstLast* CML grouping for data sets with $T = 20$ and $N = 500$, with the different tests separated by colour and the severity of the violation of the null hypothesis (split or shift of $\beta_{1,n,t}$ values between panel waves) on the x-axis. The rejection rates presented herein are based on 1 000 simulated data sets for the Lagrange multiplier type test and 100 for the Wald type test.

Table B.1
Shares of times that a particular componentwise test had the largest test statistic for simulations with an autoregressive error process, by number of individuals N, number of observations per individual T and autocorrelation coefficient ρ for the test with the *FirstLast* groupings.

N	T	test	−0.99	−0.95	−0.5	−0.2	−0.05	0	0.05	0.2	0.5	0.95	0.99
100	10	β_0	0.37	0.43	0.46	0.38	0.40	0.43	0.40	0.41	0.34	0.39	0.29
		σ_γ	0.34	0.23	0.29	0.28	0.38	0.29	0.38	0.34	0.31	0.26	0.38
		σ_ε	0.29	0.34	0.25	0.34	0.22	0.29	0.22	0.25	0.35	0.35	0.33
	14	β_0	0.42	0.40	0.44	0.40	0.45	0.43	0.46	0.43	0.37	0.41	0.33
		σ_γ	0.29	0.34	0.33	0.41	0.30	0.29	0.28	0.30	0.34	0.31	0.34
		σ_ε	0.29	0.26	0.23	0.19	0.25	0.29	0.26	0.27	0.29	0.28	0.33
	20	β_0	0.49	0.51	0.40	0.43	0.36	0.43	0.36	0.40	0.39	0.38	0.42
		σ_γ	0.29	0.20	0.32	0.29	0.37	0.29	0.31	0.34	0.30	0.34	0.28
		σ_ε	0.22	0.29	0.28	0.28	0.27	0.29	0.33	0.26	0.31	0.28	0.30
	30	β_0	0.52	0.42	0.53	0.47	0.37	0.43	0.33	0.41	0.41	0.38	0.31
		σ_γ	0.18	0.28	0.19	0.23	0.31	0.29	0.34	0.32	0.31	0.35	0.37
		σ_ε	0.30	0.30	0.28	0.30	0.32	0.29	0.33	0.27	0.28	0.27	0.32
500	10	β_0	0.39	0.50	0.46	0.38	0.38	0.43	0.44	0.40	0.34	0.46	0.42
		σ_γ	0.33	0.24	0.27	0.29	0.34	0.29	0.30	0.35	0.30	0.28	0.32
		σ_ε	0.28	0.26	0.27	0.33	0.28	0.29	0.26	0.25	0.36	0.26	0.26
	14	β_0	0.46	0.41	0.45	0.50	0.50	0.43	0.40	0.47	0.42	0.43	0.43
		σ_γ	0.26	0.26	0.30	0.28	0.29	0.29	0.35	0.29	0.33	0.27	0.27
		σ_ε	0.28	0.33	0.25	0.22	0.21	0.29	0.25	0.24	0.25	0.30	0.30
	20	β_0	0.46	0.46	0.43	0.44	0.44	0.43	0.42	0.42	0.41	0.40	0.43
		σ_γ	0.29	0.26	0.30	0.29	0.30	0.29	0.31	0.30	0.31	0.34	0.31
		σ_ε	0.25	0.28	0.27	0.27	0.26	0.29	0.27	0.28	0.28	0.26	0.26
	30	β_0	0.47	0.49	0.38	0.46	0.47	0.43	0.52	0.45	0.44	0.34	0.39
		σ_γ	0.26	0.26	0.33	0.32	0.27	0.29	0.24	0.32	0.34	0.32	0.30
		σ_ε	0.27	0.25	0.29	0.22	0.26	0.29	0.24	0.23	0.22	0.34	0.31
1000	10	β_0	0.47	0.41	0.38	0.32	0.39	0.43	0.39	0.38	0.37	0.46	0.44
		σ_γ	0.32	0.32	0.32	0.32	0.29	0.29	0.33	0.27	0.33	0.24	0.30
		σ_ε	0.21	0.27	0.30	0.36	0.32	0.29	0.28	0.35	0.30	0.30	0.26
	14	β_0	0.38	0.42	0.43	0.38	0.48	0.43	0.46	0.41	0.35	0.35	0.39
		σ_γ	0.28	0.25	0.37	0.38	0.25	0.29	0.27	0.30	0.33	0.32	0.35
		σ_ε	0.34	0.33	0.20	0.24	0.27	0.29	0.27	0.29	0.32	0.33	0.26
	20	β_0	0.37	0.44	0.45	0.41	0.45	0.43	0.48	0.42	0.38	0.42	0.42
		σ_γ	0.31	0.29	0.21	0.28	0.29	0.29	0.25	0.24	0.31	0.28	0.34
		σ_ε	0.32	0.27	0.34	0.31	0.26	0.29	0.27	0.34	0.31	0.30	0.24
	30	β_0	0.45	0.43	0.49	0.52	0.51	0.43	0.45	0.47	0.42	0.39	0.37
		σ_γ	0.26	0.30	0.27	0.32	0.24	0.29	0.26	0.26	0.31	0.28	0.32
		σ_ε	0.29	0.27	0.24	0.16	0.25	0.29	0.29	0.27	0.27	0.33	0.31

Table B.2
Shares of times that a particular componentwise test had the largest test statistic for simulations with an autoregressive error process, by number of individuals N, number of observations per individual T and autocorrelation coefficient ρ for the test with the *NearFar* groupings. Values larger or equal to 0.66 are highlighted in bold.

N	T	test	−0.99	−0.95	−0.5	−0.2	−0.05	0	0.05	0.2	0.5	0.95	0.99
100	10	β_0	0.09	0.06	0.07	0.18	0.30	0.37	0.39	0.30	0.02	0.00	0.00
		σ_γ	**0.84**	**0.87**	**0.78**	0.58	0.39	0.31	0.30	0.44	**0.72**	**0.97**	**0.96**
		σ_ε	0.07	0.07	0.15	0.24	0.31	0.32	0.31	0.26	0.26	0.03	0.04
	14	β_0	0.24	0.16	0.08	0.24	0.42	0.37	0.49	0.29	0.03	0.00	0.00
		σ_γ	0.56	**0.70**	**0.83**	0.53	0.34	0.31	0.26	0.43	**0.71**	**0.96**	**0.95**
		σ_ε	0.20	0.14	0.09	0.23	0.24	0.32	0.25	0.28	0.26	0.04	0.05
	20	β_0	0.34	0.20	0.04	0.19	0.28	0.37	0.30	0.35	0.04	0.00	0.00
		σ_γ	0.43	**0.68**	**0.92**	**0.69**	0.45	0.31	0.34	0.34	0.62	**0.99**	**0.98**
		σ_ε	0.23	0.12	0.04	0.12	0.27	0.32	0.36	0.31	0.34	0.01	0.02
	30	β_0	**0.81**	0.39	0.03	0.19	0.32	0.37	0.33	0.31	0.07	0.00	0.00
		σ_γ	0.14	0.39	**0.89**	0.52	0.36	0.31	0.30	0.30	0.65	**0.98**	**0.98**
		σ_ε	0.05	0.22	0.08	0.29	0.32	0.32	0.37	0.39	0.28	0.02	0.02
500	10	β_0	0.00	0.00	0.00	0.07	0.29	0.37	0.30	0.03	0.00	0.00	0.00
		σ_γ	**0.99**	**1.00**	**1.00**	**0.86**	0.38	0.31	0.40	**0.69**	**0.85**	**1.00**	**1.00**
		σ_ε	0.01	0.00	0.00	0.07	0.33	0.32	0.30	0.28	0.15	0.00	0.00
	14	β_0	0.01	0.00	0.00	0.07	0.29	0.37	0.39	0.07	0.00	0.00	0.00
		σ_γ	**0.98**	**1.00**	**1.00**	**0.88**	0.50	0.31	0.28	0.65	**0.85**	**1.00**	**1.00**
		σ_ε	0.01	0.00	0.00	0.05	0.21	0.32	0.33	0.28	0.15	0.00	0.00
	20	β_0	0.12	0.00	0.00	0.04	0.30	0.37	0.35	0.06	0.00	0.00	0.00
		σ_γ	**0.80**	**0.99**	**1.00**	**0.89**	0.40	0.31	0.35	0.61	**0.88**	**1.00**	**1.00**
		σ_ε	0.08	0.01	0.00	0.07	0.31	0.32	0.30	0.33	0.12	0.00	0.00
	30	β_0	**0.89**	0.18	0.00	0.03	0.31	0.37	0.39	0.05	0.00	0.00	0.00
		σ_γ	0.09	**0.68**	**1.00**	**0.91**	0.37	0.31	0.33	**0.67**	**0.87**	**1.00**	**1.00**
		σ_ε	0.02	0.14	0.00	0.06	0.32	0.32	0.28	0.28	0.13	0.00	0.00
1000	10	β_0	0.00	0.00	0.00	0.01	0.22	0.37	0.31	0.01	0.00	0.00	0.00
		σ_γ	**1.00**	**1.00**	**1.00**	**0.99**	0.44	0.31	0.37	**0.71**	**0.97**	**1.00**	**1.00**
		σ_ε	0.00	0.00	0.00	0.00	0.34	0.32	0.32	0.28	0.03	0.00	0.00
	14	β_0	0.00	0.00	0.00	0.02	0.25	0.37	0.27	0.02	0.00	0.00	0.00
		σ_γ	**1.00**	**1.00**	**1.00**	**0.96**	0.54	0.31	0.46	0.63	**0.97**	**1.00**	**1.00**
		σ_ε	0.00	0.00	0.00	0.02	0.21	0.32	0.27	0.35	0.03	0.00	0.00
	20	β_0	0.05	0.00	0.00	0.01	0.24	0.37	0.33	0.00	0.00	0.00	0.00
		σ_γ	**0.93**	**1.00**	**1.00**	**0.98**	0.50	0.31	0.39	**0.69**	**0.93**	**1.00**	**1.00**
		σ_ε	0.02	0.00	0.00	0.01	0.26	0.32	0.28	0.31	0.07	0.00	0.00
	30	β_0	**0.95**	0.07	0.00	0.00	0.24	0.37	0.39	0.01	0.00	0.00	0.00
		σ_γ	0.05	**0.87**	**1.00**	**0.98**	0.49	0.31	0.32	**0.72**	**0.93**	**1.00**	**1.00**
		σ_ε	0.00	0.06	0.00	0.02	0.27	0.32	0.29	0.27	0.07	0.00	0.00

Table B.3
Shares of times that a particular componentwise test had the largest test statistic for simulations with a shift in the $\beta_{1,n,t}$ parameter between panel waves, by number of individuals N, number of observations per individual T and size of the shift α for the test with the *FirstLast* groupings. Values larger or equal to 0.66 are highlighted in bold.

N	T	test	0	0.01	0.05	0.1	0.25	0.5	0.75	0.95	1	1.5	2
		β_0	0.33	0.35	0.31	0.15	0.03	0.00	0.00	0.00	0.00	0.00	0.00
	10	β_1	0.31	0.32	0.42	0.57	**0.95**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.36	0.33	0.27	0.28	0.02	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.32	0.32	0.30	0.14	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	14	β_1	0.35	0.32	0.44	**0.72**	**0.96**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
100		σ_γ	0.33	0.36	0.26	0.14	0.04	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.36	0.36	0.28	0.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	20	β_1	0.31	0.36	0.39	**0.70**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.33	0.28	0.33	0.17	0.01	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.33	0.28	0.19	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	30	β_1	0.31	0.39	0.53	**0.86**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.36	0.33	0.28	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.35	0.34	0.20	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	10	β_1	0.30	0.36	0.65	**0.97**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.35	0.30	0.15	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.35	0.34	0.14	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	14	β_1	0.31	0.30	**0.74**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
500		σ_γ	0.34	0.36	0.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.37	0.33	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	20	β_1	0.32	0.34	**0.78**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.32	0.33	0.13	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.33	0.23	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	30	β_1	0.31	0.46	**0.91**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.36	0.31	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.34	0.37	0.10	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	10	β_1	0.31	0.32	**0.80**	**0.98**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.35	0.31	0.10	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.36	0.32	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	14	β_1	0.33	0.39	**0.84**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
1000		σ_γ	0.31	0.29	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.32	0.29	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	20	β_1	0.41	0.35	**0.87**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.27	0.36	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		β_0	0.40	0.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	30	β_1	0.31	0.45	**0.97**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.29	0.25	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table B.4
Shares of times that a particular componentwise test had the largest test statistic for simulations with a shift in the $\beta_{1,n,t}$ parameter between panel waves, by number of individuals N, number of observations per individual T and size of the shift α for the test with the *NearFar* groupings. Values larger or equal to 0.66 are highlighted in bold.

N	T	test	0	0.01	0.05	0.1	0.25	0.5	0.75	0.95	1	1.5	2
100	10	β_0	0.30	0.33	0.36	0.34	0.31	0.35	0.33	0.31	0.30	0.30	0.33
		β_1	0.36	0.31	0.28	0.34	0.31	0.23	0.21	0.13	0.16	0.17	0.29
		σ_γ	0.34	0.36	0.36	0.32	0.38	0.42	0.46	0.56	0.54	0.53	0.38
	14	β_0	0.41	0.36	0.37	0.35	0.35	0.34	0.26	0.26	0.24	0.24	0.31
		β_1	0.27	0.29	0.33	0.32	0.31	0.26	0.17	0.16	0.15	0.16	0.18
		σ_γ	0.32	0.35	0.30	0.33	0.34	0.40	0.57	0.58	0.61	0.60	0.51
	20	β_0	0.35	0.38	0.35	0.31	0.33	0.33	0.30	0.26	0.25	0.23	0.32
		β_1	0.37	0.35	0.40	0.40	0.37	0.24	0.12	0.12	0.12	0.12	0.18
		σ_γ	0.28	0.27	0.25	0.29	0.30	0.43	0.58	0.62	0.63	0.65	0.50
	30	β_0	0.33	0.34	0.33	0.29	0.32	0.29	0.18	0.15	0.14	0.17	0.25
		β_1	0.26	0.25	0.27	0.24	0.21	0.08	0.04	0.04	0.03	0.13	0.19
		σ_γ	0.41	0.41	0.40	0.47	0.47	0.63	**0.78**	**0.81**	**0.83**	**0.70**	0.56
500	10	β_0	0.45	0.47	0.37	0.42	0.37	0.39	0.32	0.20	0.22	0.13	0.25
		β_1	0.33	0.31	0.34	0.30	0.31	0.13	0.05	0.06	0.02	0.13	0.19
		σ_γ	0.22	0.22	0.29	0.28	0.32	0.48	0.63	**0.74**	**0.76**	**0.74**	0.56
	14	β_0	0.28	0.32	0.28	0.28	0.34	0.20	0.18	0.14	0.12	0.20	0.23
		β_1	0.37	0.36	0.34	0.33	0.26	0.18	0.05	0.04	0.04	0.12	0.15
		σ_γ	0.35	0.32	0.38	0.39	0.40	0.62	**0.77**	**0.82**	**0.84**	**0.68**	0.62
	20	β_0	0.33	0.32	0.34	0.34	0.33	0.22	0.11	0.05	0.04	0.06	0.18
		β_1	0.35	0.37	0.34	0.34	0.28	0.05	0.01	0.01	0.01	0.05	0.14
		σ_γ	0.32	0.31	0.31	0.32	0.40	**0.73**	**0.88**	**0.94**	**0.94**	**0.89**	0.67
	30	β_0	0.33	0.32	0.35	0.34	0.32	0.13	0.04	0.00	0.00	0.01	0.08
		β_1	0.36	0.38	0.26	0.30	0.17	0.02	0.00	0.00	0.00	0.00	0.13
		σ_γ	0.31	0.30	0.39	0.36	0.51	**0.85**	**0.96**	**1.00**	**1.00**	**0.99**	**0.79**
1000	10	β_0	0.38	0.42	0.41	0.43	0.35	0.26	0.10	0.05	0.07	0.11	0.28
		β_1	0.29	0.30	0.33	0.30	0.24	0.07	0.01	0.02	0.01	0.07	0.11
		σ_γ	0.33	0.28	0.26	0.27	0.41	**0.67**	**0.89**	**0.93**	**0.92**	**0.82**	0.61
	14	β_0	0.31	0.32	0.26	0.29	0.36	0.20	0.08	0.03	0.04	0.07	0.19
		β_1	0.38	0.35	0.35	0.32	0.28	0.08	0.03	0.01	0.01	0.05	0.21
		σ_γ	0.31	0.33	0.39	0.39	0.36	**0.72**	**0.89**	**0.96**	**0.95**	**0.88**	0.60
	20	β_0	0.31	0.28	0.24	0.25	0.37	0.22	0.03	0.00	0.00	0.01	0.12
		β_1	0.31	0.34	0.35	0.32	0.19	0.00	0.00	0.00	0.00	0.03	0.11
		σ_γ	0.38	0.38	0.41	0.43	0.44	**0.78**	**0.97**	**1.00**	**1.00**	**0.96**	**0.77**
	30	β_0	0.36	0.31	0.34	0.30	0.25	0.04	0.00	0.00	0.00	0.00	0.06
		β_1	0.33	0.38	0.33	0.31	0.14	0.00	0.00	0.00	0.00	0.00	0.05
		σ_γ	0.31	0.31	0.33	0.39	0.61	**0.96**	**1.00**	**1.00**	**1.00**	**1.00**	**0.89**

Table B.5
Shares of times that a particular componentwise test had the largest test statistic for simulations with a split in the $\beta_{1,n,t}$ parameter in second panel wave, by number of individuals N, number of observations per individual T and size of the shift α for the test with the *FirstLast* groupings. Values larger or equal to 0.66 are highlighted in bold.

N	T	test	0	0.01	0.05	0.1	0.25	0.5	0.75	0.95	1	1.5	2
		β_0	0.33	0.34	0.35	0.35	0.38	0.35	0.18	0.07	0.08	0.01	0.03
	10	β_1	0.31	0.30	0.30	0.32	0.30	0.37	0.61	**0.77**	**0.74**	**0.90**	**0.96**
		σ_γ	0.36	0.36	0.35	0.33	0.32	0.28	0.21	0.16	0.18	0.09	0.01
		β_0	0.32	0.30	0.34	0.36	0.37	0.22	0.07	0.03	0.03	0.01	0.02
	14	β_1	0.35	0.36	0.37	0.33	0.32	0.54	**0.74**	**0.82**	**0.83**	**0.94**	**0.95**
100		σ_γ	0.33	0.34	0.29	0.31	0.31	0.24	0.19	0.15	0.14	0.05	0.03
		β_0	0.36	0.37	0.37	0.40	0.34	0.31	0.07	0.01	0.00	0.00	0.04
	20	β_1	0.31	0.32	0.32	0.27	0.28	0.48	**0.71**	**0.82**	**0.85**	**0.91**	**0.91**
		σ_γ	0.33	0.31	0.31	0.33	0.38	0.21	0.22	0.17	0.15	0.09	0.05
		β_0	0.33	0.33	0.35	0.38	0.39	0.23	0.04	0.01	0.01	0.00	0.07
	30	β_1	0.31	0.34	0.27	0.26	0.31	0.47	**0.80**	**0.80**	**0.81**	**0.86**	**0.90**
		σ_γ	0.36	0.33	0.38	0.36	0.30	0.30	0.16	0.19	0.18	0.14	0.03
		β_0	0.35	0.32	0.32	0.28	0.30	0.07	0.00	0.00	0.00	0.00	0.00
	10	β_1	0.30	0.33	0.32	0.36	0.41	0.72	**0.93**	**0.98**	**0.98**	**0.99**	**1.00**
		σ_γ	0.35	0.35	0.36	0.36	0.29	0.21	0.07	0.02	0.02	0.01	0.00
		β_0	0.35	0.37	0.36	0.37	0.27	0.05	0.00	0.00	0.00	0.00	0.00
	14	β_1	0.31	0.31	0.33	0.31	0.38	0.71	**0.94**	**0.99**	**1.00**	**1.00**	**1.00**
500		σ_γ	0.34	0.32	0.31	0.32	0.35	0.24	0.06	0.01	0.00	0.00	0.00
		β_0	0.37	0.37	0.38	0.38	0.31	0.02	0.00	0.00	0.00	0.00	0.00
	20	β_1	0.32	0.32	0.32	0.32	0.41	**0.82**	**0.94**	**0.98**	**0.98**	**1.00**	**1.00**
		σ_γ	0.32	0.31	0.30	0.30	0.28	0.16	0.06	0.02	0.02	0.00	0.00
		β_0	0.33	0.33	0.32	0.33	0.18	0.01	0.00	0.00	0.00	0.00	0.00
	30	β_1	0.31	0.31	0.30	0.28	0.56	**0.87**	**0.93**	**0.94**	**0.95**	**0.99**	**1.00**
		σ_γ	0.36	0.36	0.38	0.39	0.26	0.12	0.07	0.06	0.05	0.01	0.00
		β_0	0.34	0.33	0.30	0.35	0.26	0.01	0.00	0.00	0.00	0.00	0.00
	10	β_1	0.31	0.34	0.34	0.34	0.41	**0.81**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.35	0.33	0.36	0.31	0.33	0.18	0.01	0.00	0.00	0.00	0.00
		β_0	0.36	0.36	0.36	0.35	0.22	0.00	0.00	0.00	0.00	0.00	0.00
	14	β_1	0.33	0.33	0.31	0.32	0.48	**0.89**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**
1000		σ_γ	0.31	0.31	0.33	0.33	0.30	0.11	0.00	0.00	0.00	0.00	0.00
		β_0	0.32	0.34	0.38	0.38	0.27	0.00	0.00	0.00	0.00	0.00	0.00
	20	β_1	0.41	0.39	0.34	0.32	0.54	**0.93**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.27	0.27	0.28	0.30	0.19	0.07	0.01	0.00	0.00	0.00	0.00
		β_0	0.40	0.41	0.40	0.40	0.16	0.00	0.00	0.00	0.00	0.00	0.00
	30	β_1	0.31	0.31	0.29	0.35	0.58	**0.96**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**
		σ_γ	0.29	0.28	0.31	0.25	0.26	0.04	0.01	0.00	0.00	0.00	0.00

Table B.6
Shares of times that a particular componentwise test had the largest test statistic for simulations with a split in the $\beta_{1,n,t}$ parameter in second panel wave, by number of individuals N, number of observations per individual T and size of the shift α for the test with the *NearFar* groupings. Values larger or equal to 0.66 are highlighted in bold.

N	T	test	0	0.01	0.05	0.1	0.25	0.5	0.75	0.95	1	1.5	2
100	10	β_0	0.30	0.31	0.29	0.29	0.31	0.27	0.30	0.35	0.32	0.30	0.29
		β_1	0.36	0.36	0.36	0.37	0.35	0.35	0.38	0.33	0.34	0.40	0.44
		σ_γ	0.34	0.33	0.35	0.34	0.34	0.38	0.32	0.32	0.34	0.30	0.27
	14	β_0	0.41	0.38	0.35	0.40	0.41	0.42	0.34	0.32	0.32	0.30	0.32
		β_1	0.27	0.26	0.27	0.23	0.24	0.28	0.27	0.30	0.29	0.32	0.36
		σ_γ	0.32	0.36	0.38	0.37	0.35	0.30	0.39	0.38	0.39	0.38	0.32
	20	β_0	0.35	0.38	0.34	0.37	0.33	0.31	0.23	0.26	0.26	0.28	0.17
		β_1	0.37	0.35	0.37	0.34	0.38	0.34	0.40	0.34	0.33	0.39	0.34
		σ_γ	0.28	0.27	0.29	0.29	0.29	0.35	0.37	0.40	0.41	0.33	0.49
	30	β_0	0.33	0.34	0.37	0.30	0.31	0.30	0.35	0.32	0.31	0.26	0.21
		β_1	0.26	0.22	0.20	0.23	0.30	0.33	0.25	0.26	0.27	0.39	0.40
		σ_γ	0.41	0.44	0.43	0.47	0.39	0.37	0.40	0.42	0.42	0.35	0.39
500	10	β_0	0.45	0.48	0.39	0.40	0.34	0.39	0.37	0.32	0.30	0.18	0.26
		β_1	0.33	0.30	0.34	0.34	0.46	0.26	0.19	0.28	0.30	0.46	0.35
		σ_γ	0.22	0.22	0.27	0.26	0.20	0.35	0.44	0.40	0.40	0.36	0.39
	14	β_0	0.28	0.26	0.30	0.31	0.30	0.27	0.30	0.25	0.23	0.15	0.21
		β_1	0.37	0.38	0.35	0.35	0.32	0.24	0.26	0.32	0.30	0.58	0.43
		σ_γ	0.35	0.36	0.35	0.34	0.38	0.49	0.44	0.43	0.47	0.27	0.36
	20	β_0	0.33	0.33	0.35	0.32	0.33	0.35	0.31	0.26	0.24	0.10	0.16
		β_1	0.35	0.36	0.35	0.37	0.35	0.31	0.26	0.30	0.32	0.57	0.37
		σ_γ	0.32	0.31	0.30	0.32	0.32	0.35	0.43	0.44	0.44	0.33	0.47
	30	β_0	0.33	0.32	0.34	0.31	0.29	0.29	0.22	0.16	0.16	0.01	0.15
		β_1	0.36	0.37	0.34	0.36	0.32	0.28	0.25	0.31	0.33	0.70	0.30
		σ_γ	0.31	0.31	0.32	0.33	0.39	0.43	0.53	0.53	0.51	0.29	0.55
1000	10	β_0	0.38	0.44	0.44	0.39	0.35	0.34	0.29	0.25	0.26	0.15	0.18
		β_1	0.29	0.28	0.24	0.33	0.28	0.30	0.27	0.33	0.35	0.46	0.39
		σ_γ	0.33	0.28	0.32	0.28	0.37	0.36	0.44	0.42	0.39	0.39	0.43
	14	β_0	0.31	0.30	0.30	0.27	0.35	0.34	0.25	0.26	0.26	0.12	0.14
		β_1	0.38	0.34	0.34	0.37	0.35	0.26	0.31	0.34	0.37	0.58	0.37
		σ_γ	0.31	0.36	0.36	0.36	0.30	0.40	0.44	0.40	0.37	0.30	0.49
	20	β_0	0.31	0.30	0.27	0.27	0.32	0.43	0.29	0.27	0.21	0.05	0.12
		β_1	0.31	0.30	0.36	0.31	0.28	0.28	0.21	0.27	0.30	**0.67**	0.42
		σ_γ	0.38	0.40	0.37	0.42	0.40	0.29	0.50	0.46	0.49	0.28	0.46
	30	β_0	0.36	0.38	0.36	0.35	0.35	0.33	0.23	0.15	0.13	0.02	0.09
		β_1	0.33	0.32	0.29	0.29	0.22	0.15	0.13	0.19	0.27	**0.68**	0.26
		σ_γ	0.31	0.30	0.35	0.36	0.43	0.52	0.64	**0.66**	0.60	0.30	0.65

Table B.7
Mean estimation times in seconds for the CML estimation of models on differently sized data sets with varying sizes of T and N. Each combination based on at least 2 100 estimated models. Estimation was started from true parameter values. Standard deviations are shown in brackets below the mean values. Estimations were performed on a single processor core of either an Intel(R) Xeon(R) Bronze 3204 CPU at 1.9 GHz or an Intel(R) Xeon(R) CPU E5-2687 W v4 at 3.0 GHz.

N	$T = 10$	$T = 14$	$T = 20$	$T = 30$
100	72.99	73.02	103.48	100.72
	(10.5)	(9.87)	(16.43)	(16.58)
500	82.75	97.1	124.57	179.33
	(12.83)	(15.31)	(18.57)	(27.56)
1000	110.21	121.24	176.54	211.7
	(18.09)	(19.14)	(26.23)	(29.28)

Table B.8
Overview of required computation times in seconds to perform Lagrange multiplier and Wald type test to test for differences between the *FirstLast* groupings with shifts or splits in the parameter values between panel waves with $N = 500$ and $T = 20$. The estimation times for the Wald type test includes the time required for the estimation of the unrestricted model, starting from the parameter values of the restricted model. The quotient denotes the quotient between the time for the Wald type test and the tie for the Lagrange multiplier type test. Calculations were performed on a single processor core of an Intel(R) Xeon(R) Bronze 3204 CPU at 1.9 GHz.

Test	mean	sd	min	max
Lagrange multiplier	0.23	0.04	0.16	0.36
Wald	180.35	30.76	113.07	327.91
quotient	793.98	178.06	420.16	1593.64

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